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ZC3H11A mutations cause high myopia by triggering PI3K-AKT and NF-κB mediated signaling pathway in humans and mice

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eLife Assessment

This work investigates ZC3H11A as a cause of high myopia through the analysis of human data and experiments with genetic knockout of Zc3h11a in mouse, providing a **useful** model of myopia. The evidence supporting the conclusion is still **incomplete** in the revised manuscript as the concerns raised in the previous review were not fully addressed. The article would benefit from a more robust genetic analysis and comprehensive presentation of human phenotypic data to clarify the modes of inheritance in the families, currently limited by loss of patient follow-up and addressing whether there is a reduction in bipolar cell number or decreased marker protein expression through cell counts or quantifiable, less saturated Western blots. The work will be of interest to ophthalmologists and researchers working on myopia

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Abstract

High myopia (HM) is a severe form of refractive error that results in irreversible visual impairment and even blindness. However, the genetic and pathological mechanisms underlying this condition are not yet fully understood. From a cohort of 1015 patients with high myopia in adolescents, likely pathogenic missense mutations were identified in the *ZC3H11A* gene in four patients by whole exome sequencing. This gene is a zinc finger and stress-induced protein that plays a significant role in regulating nuclear mRNA export. To better understand the function and molecular pathogenesis of myopia in relation to gene mutations, a *Zc3h11a* knock-out (KO) mouse model was created. The heterozygous KO (Het-KO) mice exhibited significant shifts in refraction towards myopia. Myopia-related factors, including TGF-β1, MMP-2 and IL-6 were found to be upregulated in the retina or sclera, and electroretinography and immunofluorescence staining results showed dysfunction and reduced number of bipolar cells in the retina. Transmission electron microscopy findings

suggesting ultrastructural abnormalities of the retina and sclera. Retinal transcriptome sequencing showed that 769 genes were differentially expressed, and *Zc3h11a* was found to have a negative impact on the PI3K-AKT and NF-κB signaling pathways by quantitative PCR and western blotting. In summary, this study characterized a new candidate pathogenic gene associated with high myopia, and indicated that the ZC3H11A protein may serve as a stress-induced nuclear response trigger, and its abnormality causes disturbances in a series of inflammatory and myopic factors. These findings offer potential therapeutic intervention targets for controlling the development of HM.

Introduction

Myopia is a highly prevalent eye affliction that commonly develops during childhood and early adolescence. The prevalence of myopia among the adult population ranges from 10-30%, while parts of East and Southeast Asia have reported rates as high as 80- 90% among young people (Baird et al., 2020; Holden et al., 2016). HM is a severe refractive error characterised by a diopter ≤ -6.00 D or an axial length greater than 26 mm. It is estimated that by 2050, the number of people with HM worldwide will reach 938 million, accounting for 9.8% of the total population. HM can trigger a range of adverse ocular changes, such as cataracts, glaucoma, retinal detachment, macular degeneration and possibly total blindness (Koga et al., 2014; Saw et al., 2005). While conventional methods for the prevention and control of myopia can provide some correction, they are not entirely effective in managing its progression, particularly during childhood and adolescence.

With the development of next-generation sequencing, whole exome sequencing (WES) and whole genome sequencing (WGS) have extended the findings of linkage studies to identify potential causes of syndromic HM (sHM) and non-syndromic HM (nsHM). To date, approximately 20 genes with causal associations have been identified in sHM, including *ZNF644, SCO2, CCDC111, LRPAP1, SLC39A5, LEPREL1, P4HA2, OPN1LW, ARR3, BSG, NDUFAF7, CPSF1, TNFRSF21, DZIP1, XYLT1, CTSH, GRM6, LOXL3* and *GLRA2* (Haarman et al., 2022; Tian et al., 2023; Yang et al., 2023; Ye et al., 2023). These discoveries have provided insight into the molecular mechanisms underlying HM. However, known candidate genes can only explain about 20% of the causes of this disease (Cai et al., 2019; Tedja et al., 2019). At the same time, the neuromodulators and signal molecules of HM are extremely complex, including sclera extracellular matrix (ECM) remodelling and endoplasmic reticulum (ER) stress (Ikeda et al., 2022), inflammatory responses, the release of dopamine and gamma- aminobutyric acid (GABA), or abnormalities in myopia-related signaling pathways, such as retinoic acid signaling, TGF-β signaling and HIF-1α signaling.

Screening mutations from an in-house adolescents high myopia survey cohort by WES, we identified zinc finger CCCH domain-containing protein 11A (*ZC3H11A*) as a HM candidate gene. This particular gene is a member of the zinc finger protein gene family. Multiple zinc finger protein genes (e.g., *ZNF644*, *ZC3H11B*, *ZFP161*, *ZENK*) are associated with myopia or HM. Of these, *ZC3H11B* (a human homolog of *ZC3H11A*) and five GWAS loci (Schippert et al., 2007; Shi et al., 2011; Szczerkowska et al., 2019; Tang et al., 2020; Wang et al., 2004) correlate with AL elongation or HM severity. Proteomic studies further suggest ZC3H11A involvement in the TREX complex, implicating RNA export mechanisms in myopia pathogenesis. Additionally, it has been suggested that this protein is involved in stress-induced responses (Younis et al., 2018). Dysfunction of *ZC3H11A* results in enhanced NF-κB signaling through defective IκBα protein expression, which is accompanied by upregulation of numerous innate immune- and inflammation-related mRNAs, including *IL-8, IL-6* and *TNF* in vitro (Jimi et al., 2019). Moreover, patients with myopia have a higher proportion of inflammation- associated cells, including neutrophils, while those with moderate myopia show strained immune system function (Lin et al., 2016; Qi et al., 2022). Conversely, some traditional Chinese medicines can control myopia progression by suppressing AKT and NF-κB mediated inflammatory reactions (Chen et al., 2022). However, the precise pathological mechanism of *ZC3H11A* in myopia development remains unclear, necessitating further research.

The current study identified four variants in the *ZC3H11A* gene among a high myopia adolescent cohort of 1015 adolescents. Additionally, heterozygous knockout (Het-KO) mice were constructed using the CRISPR/Cas9 system and their myopic phenotypes, visual function, bipolar cell apoptosis, and retinal and scleral microstructure were assessed. Moreover, RNA sequencing and expression experiment was performed to identify perturbed molecules and pathways and to examine the interactions between these factors in relation to HM. The above results will provide new ideas for the prevention and control of high myopia, especially early-onset, uncorrectable and familial myopia.

Results

ZC3H11A mutations are associated with HM in a Chinese cohort

The genomic data is sourced from an independently established genetic cohort of high myopia patients. Four missense mutations in the *ZC3H11A* gene (c.412G>A, p.V138I; c.128G>A, p.G43E; c.461C>T, p.P154L; and c.2239T>A, p.S747T) were identified in the 1015 HM patients aged from 15 to 18 years. The uncorrected visual acuity (UCVA) and axial length of these patients are presented in **Table 1**. All of the identified mutations exhibited very low frequencies in the Genome Aggregation Database (gnomAD) and Clinvar, and using pathogenicity prediction software SIFT, PolyPhen2, and CADD, most of them display high pathogenicity levels. Among them, c.412G>A, c.128G>A and c.461C>T were located in or around a domain named zf-CCCH_3 (**Figure 1A and B**). Furthermore, all of the mutation sites were located in highly conserved amino acids across different species (**Figure 1C**). Four mutations resulted in a higher degree of conformational flexibility and altered the negative charge at the corresponding sites (**Figure 1D and E**).

Zc3h11a Het-KO mice exhibited myopic phenotypes

Het-KO mice were constructed at four weeks of age against a C57BL/6J background using CRISPR/Cas9 technology (**Supplement Figure 1**). Retinal fundus images and ocular histomorphology of *Zc3h11a* Het-KO mice at eight weeks postnatal were assessed against those of their wild-type (WT) counterparts. No significant structural differences were observed (**Supplement Figure 2**). These findings suggest that the deletion of *Zc3h11a* does not alter the retinal structure or ocular histomorphology. Therefore, *Zc3h11a* Het-KO mice are a relevant model for the investigation of the role of *Zc3h11a* in refractive development.

Refraction and axial length in *Zc3h11a* Het-KO mice were found to be significantly greater than in WT littermates (independent samples t-test, $p<0.05$; **Figure 2A and B**). The difference between the two genotypes was statistically significant at weeks 4 and 6. Correspondingly, the vitreous chamber depth of *Zc3h11a* Het-KO mice was deeper than that of WT littermates (independent samples t-test, $p<0.05$; **Figure 2C**). There were no significant differences in the anterior chamber depth, lens diameter and body weight between the two groups (**Figure 2D-F**). These results suggest that the Het-KO of the *Zc3h11a* gene can lead to an increase in myopia in mice.

Figure 1.

Structural and evolutionary analysis of *ZC3H11A* mutations in high myopia.

(A) Enomic organization of *ZC3H11A*: Exon-intron structure (exons 1-20) with four identified missense mutations (orange arrows: c.128G>A/p.G43E, c.412G>A/p.V138I, c.461C>T/p.P154L, c.2239T>A/p.S747T). Domain architecture showing the zf-CCCH_3 zinc finger domain (exons 5-8) harboring three mutations (G43E, V138I, P154L). (B) Full-length ZC3H11A structural model: Predicted tertiary structure (PyMOL v2.5) with mutation sites highlighted: G43E (exon 6, orange), V138I/P154L (exon 8, green and blue), S747T (exon 20, purple). (C) Mutation localization: Schematic mapping of mutations to exons: G43E (exon 6), V138I/P154L (exon 8), S747T (exon 20). Exon sizes scaled proportionally. (D) Cross-species conservation: Multiple sequence alignment of *ZC3H11A* orthologs showing absolute conservation of mutated residues. (E) DynaMut2-predicted conformational flexibility changes: These mutations can result in a higher degree of conformational flexibility at the corresponding sites, potentially destabilizing the structural domain.

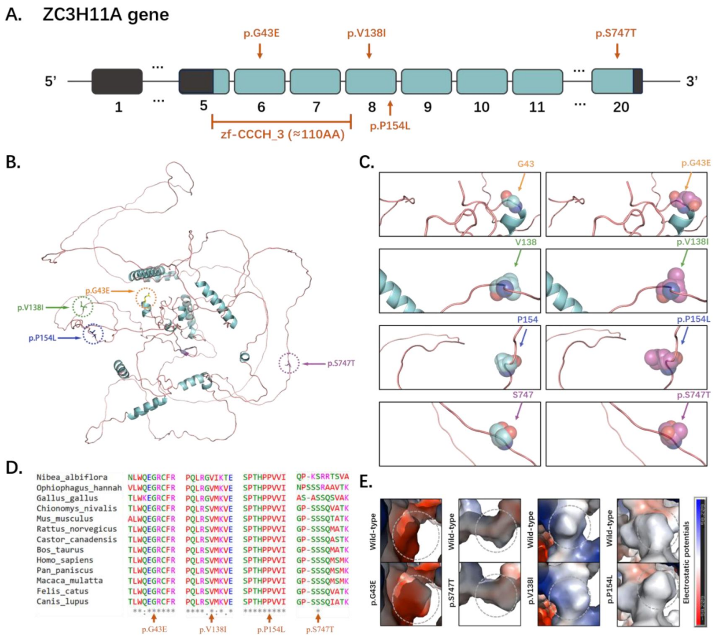


Table 1

The clinical features and mutations of affected patients

Patient information			Refraction (D)		Axial length (mm)		Variation information			Prediction software and databases				
ID.	Sex	Age	OD	OS	OD	OS	Genotype	chromosomal positions	Existing_variation (rs numbers)	SIFT (score)	PolyPhen2 (score)	CADD (score)	gnomAD	Clinvar
1	M	18	-6.50	-5.125	25.68	25.34	c.412G>A:p.Val138Ile, Het	chr1:203798692-203798692	rs142418357	D (0.03)	PB (0.912)	24.3	7.95E-06	-
2	F	15	-4.25	-6.125	24.61	25.11	c.128G>A:p.Gly43Glu, Het	chr1:203787771-203787771	-	D (0)	PS (0.906)	26	-	-
3	M	17	-8.00	-8.75	26.16	27.02	c.461C>T :p.Pro154Leu, Het	chr1:203798741-203798741	-	D (0)	PS (0.628)	29.3	-	-
4	F	15	-6.25	-3.70	25.53	24.86	c.2239T>A:p.Ser747Thr, Het	chr1:203821333-203821333	-	T (0.1)	PS (0.838)	22.7	-	-

M, male; F, female; Het, heterozygote; OD, ocular dexter; OS, oculus sinister; D, deleterious, T, tolerated; PB, probably_damaging; PS, possibly_damaging; gnomAD, Genome Aggregation Database; AF, Allele Frequency; -, Not Applicable

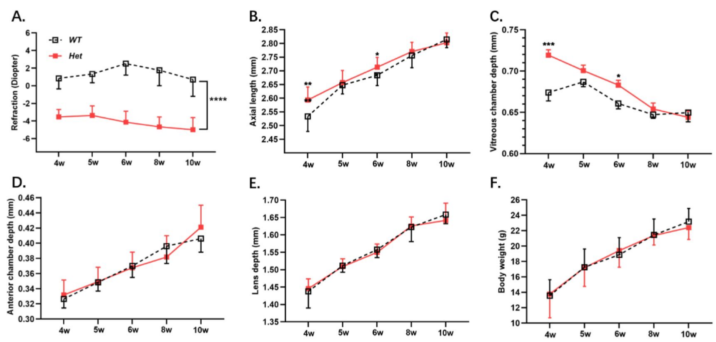


Figure 2.

Zc3h11a Het-KO mice exhibit myopic shifts in refractive parameters.

(A) Het-KO mice showed myopia in diopter. (B, C) Axial length and Vitreous chamber depth increased elongation in Het-KO mice at weeks 4 and 6. (D-F) No genotype-dependent differences in anterior chamber depth, lens thickness, or body weight. The effect of genotype on time-dependent refractive development was assessed through independent samples t-tests. P-values are indicated as follows: *P<0.05, **P<0.01, ***P<0.001 and ****<0.0001.

Reduced b-wave amplitude and bipolar cell-labelled protein abundance in *Zc3h11a* Het-KO mice

To confirm if *Zc3h11a* is responsible for refractive development regulation, visual function was assessed by electroretinography (ERG). Upon dark adaptation, b-wave amplitudes in seven-week-old Het-KO mice were significantly lower at dark 3.0 (0.48 log cd·s/m^2) and dark 10.0 (0.98 log cd·s/m^2) compared to WT mice (**Figure 3A and C**). On the contrary, there were no differences in a-wave amplitudes between the Het-KO and WT groups (**Figure 3B**). Based on the ERG results, variations between Het-KO and WT littermates can be observed. Under the dark adaptation conditions of 3.0 and 10.0, there was a significant change in the amplitude of ERG b-waves, indicating impaired bipolar cell function. Immunofluorescence analyses were performed on frozen retinal sections to investigate this phenomenon further. Specifically, Zc3h11a, PKCα, Opsin-1 and Rhodopsin markers were utilised to detect the Zc3h11a protein, rod-bipolar cells, cone cells and rod cells, respectively (**Figure 3D-F**). Quantitative analysis of immunofluorescence staining results revealed that the protein abundance of Zc3h11a and PKCα was significantly reduced in the Het-KO group (**Figure 3G**, H). Western blot analysis further showed that the expression level of PKC-α protein in the retina of Het- KO mice was significantly reduced, indicating a decrease in the number of bipolar cells (**Figure 3I**, J). However, there were no significant differences in the protein abundance of Opsin-1 and Rhodopsin between the two groups (**Supplement Figure 3**). Thus, both the function and number of retinal bipolar cells were decreased in *Zc3h11a* Het-KO mice.

Retinal ultrastructure alterations in *Zc3h11a* Het-KO mice model

Transmission electron microscopy (TEM) was used to analyse retinal ultrastructure of in 10-week-old mice. In the inner nuclear layer (INL) of the retina, in contrast to WT mice, Het-KO mouse cells had enlarged perinuclear gaps (black arrow), perinuclear cytoplasmic oedema (blue arrow), and thinned and lightened cytoplasm (**Figure 4A and B**). However, in the outer nuclear layer (ONL), no significant difference in cell morphology was observed between the two groups of mice (**Figure 4C and D**). These findings indicate structural damage to bipolar cells in Het-KO mice. Furthermore, when compared to the WT group, the Het-KO mice exhibited relatively damaged photoreceptor cell membrane discs (MB), in which the outer layer was detached and sparsely distributed locally (**Figure 4E and F**). There were also a small number of broken membrane discs, as well as some disorganized and loosely arranged ones (red arrow), with a slight increase in size (**Figure 4G and H**). This suggests impaired light signal capture, transduction, and compromised visual function in Het-KO mice.

RNA-Seq analysis of molecular and pathways changes in *Zc3h11a* Het-KO mice retinas

In the retina transcriptome analysis, 769 genes were differentially expressed (Fold change (FC) of at least two and a P value < 0.05) in the *Zc3h11a* Het-KO group, of which, 303 were upregulated and 466 were downregulated (**Figure 5A**). GO enrichment analysis revealed significant enrichment of differentially expressed genes in the following functions: Zinc ion transmembrane transport (GO:0071577) within metal ion homeostasis, associated with retinal photoreceptor maintenance (Ugarte and Osborne, 2001), RNA biosynthesis and metabolism (GO:0006366) in transcriptional regulation, potentially influencing ocular development, negative regulation of NF-κB signaling (GO:0043124) in inflammatory modulation, a pathway involved in scleral remodelling (Xiao et al., 2025), calcium ion binding (GO:0005509), critical for phototransduction (Krizaj and Copenhagen, 2002), zinc ion transmembrane transporter activity (GO:0005385), participating in retinal zinc homeostasis (**Figure 5B and C**). KEGG pathway enrichment analysis indicated that

Figure 3.

Zc3h11a Het-KO mice exhibited bipolar cell dysfunction, accompanied by reduced abundance of the key bipolar cell marker protein PKCα.

(A) Representative scotopic ERG responses from Het-KO and WT eyes at dark 0.01(2.02 log cd·s/m^2), dark 3.0 (0.48 log cd·s/$m^2$and 10.0 (0.98 log cd·s/m^2). (B, C) Quantification of a-wave (photoreceptor function) and b-wave (bipolar cell function) amplitudes. Het-KO mice (n=12) show significant b-wave reduction at dark 3.0 and dark 10.0 vs. WT (n=12). No a-wave differences observed (p>0.1). (D-F) Immunofluorescence-stained samples to detect Zc3h11a, PKCα (key bipolar cell marker protein), Opsin-1 (key cone photoreceptors marker protein) and Rhodopsin (key rod photoreceptors marker protein). (G, H) Zc3h11a and PKCα protein abundance reduced in the retina of Het-KO mice. (I) Western blot analysis showed that the PKCα protein content in the retina of Het-KO mice was decreased. (J) No differences in Opsin-1 or Rhodopsin protein abundance. Statistical significance was defined as *P < 0.05, **P<0.01 and ***P<0.001, as determined by independent samples t-tests.

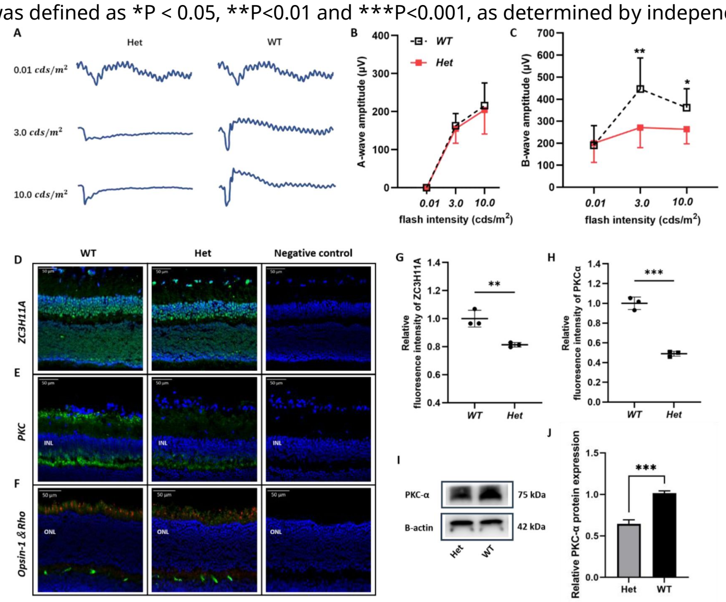
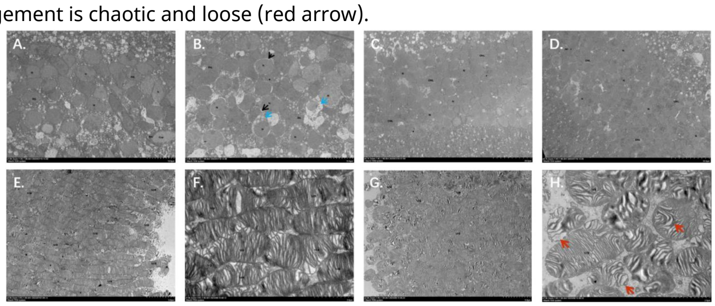


Figure 4.

TEM showing the retinal abnormalities ultrastructure of *Zc3h11a* Het-KO mice.

(A-B) Inner nuclear layer (INL). (A) WT mice, normal bipolar cell morphology; (B) *Zc3h11a* Het-KO pathological features include enlarged perinuclear gaps (black arrowheads), cytoplasmic edema (blue arrowheads), thinned/lightened cytoplasm in bipolar cells. (C-D) Outer nuclear layer (ONL). (C) WT mice and (D) *Zc3h11a* Het-KO mice Het-KO, no significant differences in photoreceptor nucleus morphology. (E-H) Photoreceptor membrane discs (MB). (E-F) WT mice, tightly stacked, uniformly arranged membrane discs. (G-H) *Zc3h11a* Het-KO structural disruptions include the outer layer falls off, the local distribution is sparse, and the arrangement is chaotic and loose (red arrow).



the differentially expression genes (DEGs) in the *Zc3h11a* Het-KO group were primarily involved in the PI3K-AKT signaling pathway, MAPK signaling pathway, and neuropsychiatric disease (**Figure 5D**). These findings preliminarily suggest dysregulated NF-κB and PI3K-AKT signaling pathways in the retinas of *Zc3h11a* Het-KO mice.

Zc3h11a negatively regulates the PI3K-AKT and NF-κB pathways

PI3K-AKT is one of the most important pathways for cell survival, division, autophagy and differentiation (Alzahrani, 2019). Meanwhile, it has been found that *ZC3H11A* can regulate the NF-κB pathway at the level of *IκBα* mRNA export in ZC3-KO cells (Darweesh et al., 2022). IκBα is a critical protein that governs NF-κB function predominantly within the cytoplasmic compartment. It plays a pivotal role in inhibiting the activation of NF-κB (Dyson and Komives, 2012; Manavalan et al., 2010). In addition, AKT promotes IκBα phosphorylation to undergo degradation, which enhances the nuclear translocation of NF-kB (Alzahrani, 2019; Torrealba et al., 2020). Based on RNA sequencing and the aforementioned literature evidence, this study investigated the regulatory role of *Zc3h11a* in the PI3K-AKT and NF-κB pathways within the mouse retina. To this end, we evaluated the mRNA and protein expression levels of Zc3h11a, PI3K, AKT, p-AKT, IκBα and NF-κB in the retina. The expression levels ofPI3K, AKT, p-AKT and NF-κB were significantly upregulated, while Zc3h11a and IκBα was downregulated in Het-KO mice (**Figure 6A-L**). Meanwhile, through transfection of overexpression mutant plasmids, it was found that compared to the wild-type, the mRNA expression levels of *IκBα* in the nucleus of all four mutant types (*ZC3H11A*^{V138I}, *ZC3H11A*^{G43E}, *ZC3H11A*^{P154L} and *ZC3H11A*^{S747T}) were significantly reduced (**Supplement Figure 4**). These results strongly suggest that *ZC3H11A* likely contributes to myopia pathogenesis by inhibiting both PI3K-AKT and NF-κB signaling pathways.

TGF-β1, MMP-2 and IL-6 were increased in *Zc3h11a* Het-KO mice

In animal models of myopia, it has been established that intraocular concentrations of TGF-β1 are elevated (Chen et al., 2013; Liu et al., 2022), especially in the retina and sclera. The abnormal expression of matrix metalloproteinase-2 (MMP-2) and interleukin-6 (IL-6) can induce myopia, and these are regulated by NF-κB (Libermann and Baltimore, 1990; Wu and Schmid-Schönbein, 2011). qPCR analysis showed increased expression of *TGF-β1*, *MMP-2*, and *IL-6* in the sclera and retina of Het-KO mice compared with WT mice (**Figure 7A and B**). The western blot results showed increased expression levels of TGF-β1 and MMP-2 in the retinas of *Zc3h11a* Het-KO mice (**Figure 7D**).Scleral TEM results suggested that scleral collagen fibres in Het-KO mice were disorganized over a large area with irregular transverse and longitudinal arrangement (**Figure 7C**). These results further corroborate that Zc3h11a Het-KO mice exhibit a myopic phenotype, and that Zc3h11a mutation may participate in the development of myopia by inhibiting the NF-κB signaling pathway, thereby upregulating the expression of myopia-related factors such as MMP-2, and IL-6.

Discussion

This study identified and validated a new candidate gene in a large high myopia cohort, *ZC3H11A*. Moreover, the study provided evidence of the presence of moderate or HM phenotypes and damaged bipolar cells *in vivo* by constructing Het-KO mice for the first time. TEM analysis showed ultrastructural changes in parts of both the retina and sclera of Het-KO mice. To understand how *ZC3H11A* regulates the development of HM, retinal transcriptome sequencing was performed. The results revealed that *Zc3h11a* mutation upregulates the PI3K-AKT and NF-κB signaling pathways. In addition, the expression levels of myopia-related factors, such as TGF-β1, MMP-2 and IL-6, were

Figure 5.

RNA-Seq reveals pathway dysregulation in the retina of *Zc3h11a* Het-KO mice.

(A) Volcano plot of differentially expressed genes (DEGs): 769 total (303 upregulated, 466 downregulated in Het-KO vs. WT retinas. (B, C) Gene Ontology (GO) enrichment analysis: Biological Process: Zinc ion transmembrane transport (GO: 0071577, photoreceptor maintenance), Negative regulation of NF-κB signaling (GO: 0043124, scleral remodeling). Molecular Function: Calcium ion binding (GO: 0005509, phototransduction), Zinc ion transmembrane transporter activity (GO: 0005385, retinal zinc homeostasis). (D) KEGG pathway enrichment analysis: Key pathways: PI3K-AKT signaling, MAPK signaling.

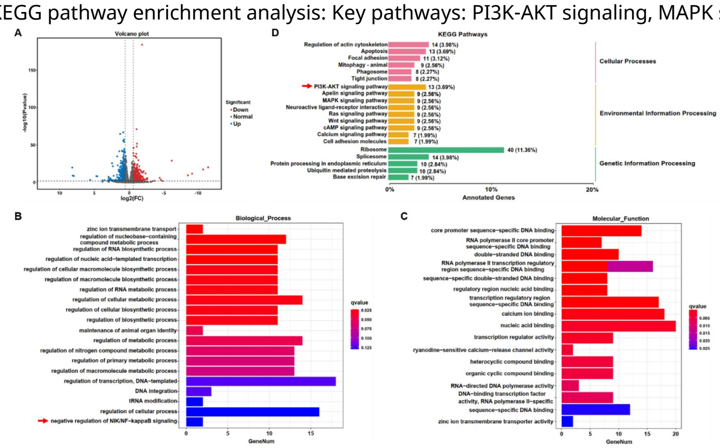
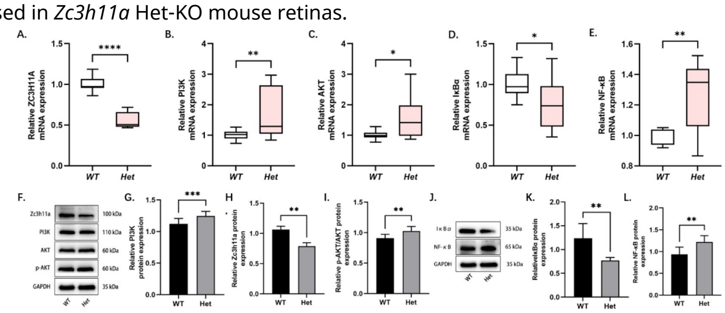


Figure 6.

Dysregulation of PI3K-AKT and NF-κB signaling pathways in the retinas of *Zc3h11a* heterozygous knockout (Het-KO) mice.

(A-E) qRT-PCR quantification of *ZC3H11A, PI3K, AKT, IκBα*, and *NF-κB* mRNA in the retina (n=3 mice/group): *Zc3h11a* expression was decreased in *Zc3h11a* Het-KO mouse retinas, while *PI3K* and *AKT* expression were increased; *IκBα* expression was decreased and *NF-κB* expression was increased. (F-L) Western blot analysis of Zc3h11a, PI3K, p-AKT/AKT, IκBα, and NF-κB protein levels in the retina (n=3 mice/group). (F-I) Quantitative analysis of ZC3H11A and PI3K levels normalized to GAPDH, while p-AKT levels were normalized to AKT: ZC3H11A expression was decreased in *Zc3h11a* Het-KO mouse retinas, while PI3K and p-AKT/AKT expression were increased. (J-L) Quantitative analyses of IκBα and NF-κB normalized to GAPDH: IκBα expression was decreased in *Zc3h11a* Het-KO mouse retinas.



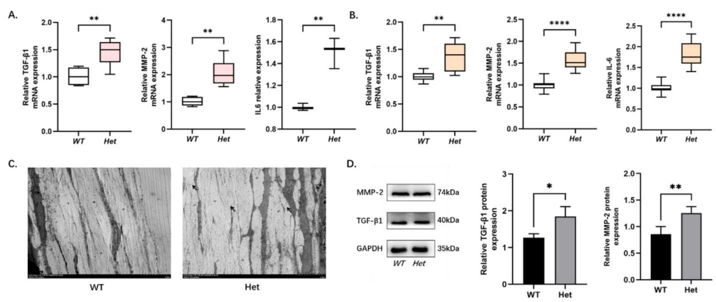


Figure 7.

Elevated expression of TGF-β1, MMP-2, and IL-6 in retina and sclera of *Zc3h11a* Het-KO mice, with disrupted scleral ultrastructure.

(A, B) qRT-PCR quantification of TGF-β1, MMP-2 and IL-6 mRNA in retina (A) and sclera (B) (n=3 mice/group): Expression of TGF-β1, MMP-2 and IL-6 was increased in both retina and sclera of Zc3h11a Het-KO mice. (C) TEM structure of sclera, WT mice, organized collagen fibers with regular transverse/longitudinal arrangement. *Zc3h11a* Het-KO mice, disorganized collagen fibers structure with irregular arrangement (black arrows). (D) Quantitative analyses of TGF-β1 and MMP-2 normalized to GAPDH (n=3 mice/group): Expression of TGF-β1 and MMP-2 was increased in retinas of Zc3h11a Het-KO mice.

upregulated in the retina and sclera of the Het-KO mice. Therefore, it can be speculated that the ZC3H11A protein may act as a stress-induced nuclear response trigger, its abnormal expression contributing to the early onset of HM.

ZC3H11A is involved in the export and post-transcriptional regulation of selected mRNA transcripts required to maintain metabolic processes in embryonic cells, these are essential for the viability of early mouse embryos (Younis et al., 2023). In studies of chickens, mice and humans, the zinc finger protein *ZENK* was found to play a role in the development of refractive myopic excursion and lengthening of the ocular axis. Moreover, early growth response gene type 1 Egr-1 (the human homolog of *ZENK*) activates the TGF-β1 gene by binding to its promoter, which is thought to be associated with myopia (Baron et al., 2006; Xiao et al., 2022). Another zinc protein finger protein 644 isoform, *ZNF644*, has recently been identified by WES as causing HM in Han Chinese families (Shi et al., 2011). *ZC3H11A* is also a zinc finger protein, and the findings of the current study revealed that *Zc3h11a* Het-KO mice showed a myopic shift, which is consistent with previous studies reporting reduced protein expression of *Zc3h11a* in an unilateral induced myopic mouse model (Fan et al., 2012).

In vertebrate models, refractive development and ocular axial growth are visually controlled (Wallman and Winawer, 2004). The regulation of axial length or refractive error occurs through complex light-dependent retina-to-sclera signaling (Tkatchenko and Tkatchenko, 2019). Optical scatter information is processed by the retina and then converted into molecular signals that regulate peripheral retinal growth and scleral connective tissue renewal, ultimately affecting the growth rate of the posterior segment of the eye (Harper and Summers, 2015; Tkatchenko et al., 2006; Tkatchenko et al., 2018). All cell types of the retina contain myopia-related genes and retinal circuitry driving refractive error. In this study, Het-KO mice were found to have reduced b-wave amplitudes, diminished the abundance of PKCα and ultrastructural changes in inner nuclear layer (INL) of the retina. Thus, damage to retinal bipolar cells may impair visual signal processing and transduction through neural circuit alterations, thereby contributing to the pathogenesis of HM.

Transcriptome sequencing and quantitative analyses collectively demonstrated downregulated PI3K-AKT and NF-κB signaling pathways in the retinas of Het-KO mice. Given the well-established dysregulation of these pathways in high myopia (Lin et al., 2016), we aimed to investigate how *Zc3h11a* mediates alterations in these pathways and their potential regulatory cross-talk in the mouse retina. Previous research revealed that a decrease in *Zc3h11a* inhibited the translocation of *IκBα* from the nucleus to the cytoplasm (Darweesh et al., 2022). *IκBα* is a downstream factor of PI3K-AKT and binds to NF-κB (p65) in the cytoplasm, inhibiting its nuclear translocation. The qPCR and western Blot results verified that *Zc3h11a* mutation negatively regulates both the PI3K-AKT and NF-κB signaling pathways. Thus, *Zc3h11a* may exert an influence on PI3K-AKT and NF-κB signaling pathways by modulating the cytoplasmic levels of IκBα, contributing to the development of myopia.

NF-κB was first discovered 25 years ago and described as a key regulator of induced gene expression in the immune system, playing a central role in the coordinated control of intrinsic immune and inflammatory responses (Hayden and Ghosh, 2011; Morgan and Liu, 2011). The downstream factors of this pathway include IL-6, IL-8, TNF-α, MMP-2, TGF-β1, etc (Jimi et al., 2019; Yoshida and Whitsett, 2006). Studies have demonstrated that the MMP-2 and IL-6 expression levels are increased in myopic eyes and that inhibiting MMP-2 or IL-6 expression will provide some degree of control over myopia progression (Lin et al., 2016; Zhao et al., 2018). Identification of the PI3K-AKT-NF-κB signaling pathway and downstream myopia effect in our study may open up new opportunities for the prevention and treatment of HM and fundus lesions in the future.

There are a number of limitations of this study that should be acknowledged. First, the patient's ophthalmic examination was not comprehensive enough, and genetic analysis such as segregation analysis was not performed. Second, Zc3h11a homozygous KO (Homo-KO) mice were not obtained in our study because homozygous deletion of exons confer embryonic lethality (Younis et al., 2023). Third, the axial length of Zc3h11a Het-KO mice was longer only at 4 and 6 weeks of age. This may be due to the small size of mouse eyes (1D refractive change corresponds to only 5-6μm axial length change) (Schaeffel., 2004) and the theoretical resolution limit of 6μm of the SD-OCT equipment used in this study (Zhou., 2008b). Fourth, while the current study observed damage to bipolar cells in Het-KO mice, an in-depth exploration of the underlying mechanism was beyond the scope of the research. Finally, there was a limited observation time in this study and the effect of *ZC3H11A* on the late refractive system was not assessed.

Overall, this study has provided four key findings. First, it was confirmed that the *ZC3H11A* is a new candidate gene associated with HM in a cohort of Chinese Han individuals. Secondly, changes in *Zc3h11a* expression were found to affect retinal function, particularly in bipolar cells. Third, changes in *Zc3h11a* expression were found to cause alterations in numerous signaling pathways, the most notable being the PI3K- AKT and NF-κB pathways. Fourth, changes in *Zc3h11a* expression were found to cause changes in the myopia-related genes TGF-β1, MMP-2 and IL-6. The above results suggest that abnormal expression of *Zc3h11a* triggers nuclear translocation defects of *IκBα*, thereby exerting negative feedback regulation of the PI3K-AKT signaling pathway and inhibiting the NF-κB signaling pathway. At the same time, the increase in TGF-β1 in myopic eyes stimulates the PI3K-AKT signaling pathway, which leads to activation of the PI3K-AKT signaling pathway, resulting in downstream degradation of IκBα after phosphorylation. This, in turn, increases the nuclear translocation of NF-κB. To sum up, *Zc3h11a* promotes the myopia-associated factor TGF-β1 by acting directly or indirectly on both the PI3K-AKT and NF-κB signaling pathway-mediated stress reactions and the expression of MMP-2 and IL-6, which together, caused the development of early high myopia (**Figure 8**). Therefore, this model can be used as one of the new strategies for intervention and treatment of high myopia. However, because the causes of myopia or HM are complex and involve different tissues and molecular pathways in the eye, it is likely that future research will identify more genes and molecular mechanisms that could provide guidance for clinical intervention and treatment.

Materials and methods

Ethics approval and consent to participate

The studies that involved human participants were approved by the Eye Hospital, Wenzhou Medical University (Wenzhou, China), and were carried out in strict adherence to the guidelines of the Helsinki Declaration. All participants provided informed consent. Furthermore, all animal experiments were approved by the Animal Care and Ethics Committee at Wenzhou Medical University and were performed according to the guidelines set forth in the Association for Research in Vision and Ophthalmology Statement for the Use of Animals in Ophthalmic and Visual Research. (*https://www.arvo.org/About/policies/statement-for-the-use-of-animals-in-ophthalmic-and-vision-research/*).

Recruitment of subjects

Four sporadic HM patients (aged from 15-18) were recruited from the Eye Hospital of Wenzhou Medical University, three of them come from the Myopia Associated Genetics and Intervention Consortium (MAGIC) project. HM was defined as SE ≤ -6.00 D in either eye (Flitcroft et al., 2019). Partial patients underwent certain ophthalmic examinations, including visual acuity (at least three measurements under non-cycloplegia), axial length and fundus photography were performed.

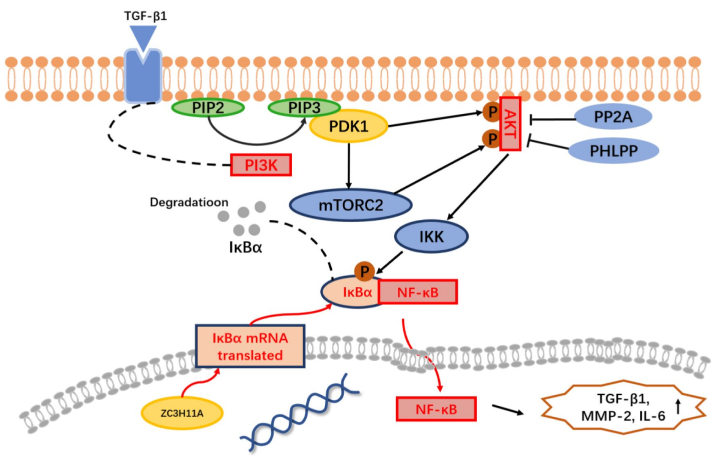


Figure 8.

Mechanism of *Zc3h11a* haploinsufficiency-driven dysregulation of PI3K-AKT/NF-κB signaling in myopia pathogenesis.

NF-κB activation, loss of *ZC3H11A* impairs nuclear export of IκBα mRNA, leading to reduced cytoplasmic IκBα protein levels and consequent hyperactivation of NF-κB. PI3K-AKT activation. *ZC3H11A* deficiency upregulates PI3K, enhancing the conversion of PIP2 to PIP3, which drives AKT phosphorylation (p-AKT). Activated p-AKT promotes IκBα degradation, further amplifying NF-κB nuclear translocation and transcriptional activity. Downstream effects, the combined hyperactivity of PI3K-AKT and NF-κB pathways elevates the expression of: TGF-β1 (extracellular matrix remodeling), MMP-2 (collagen degradation), IL-6 (pro-inflammatory signaling), collectively driving scleral thinning and inflammatory responses in myopia pathogenesis.

WES and the detection of variants

DNA was extracted from all probands using a FlexiGene DNA Kit (Qiagen, Venlo, The Netherlands), according to the manufacturer's protocol. DNA from all probands underwent WES using a Twist Human Core Exome Kit and an Illumina NovaSeq 6000 sequencing system (150PE) (Berry Genomics Institute, Beijing, China). The mean depth and coverage of the target region were approximately 78.26× and 99.7%, respectively. Sequence reads were aligned to the reference human genome (UCSC hg19) using the Burrows-Wheeler aligner (BWA). The variants were called and annotated with Verita Trekker and Enliven software (Transcript ID, ENST00000332127.4, Berrry Genomics Institute), respectively.

Het-KO mice model construction, husbandry and ocular biometric measurements

The germline heterozygous *Zc3h11a* knockout (Het-KO) mice were generated by CRISPR/Cas9-mediated gene editing at the embryonic stage on a C57BL/6J background, provided by GemPharmatech Co., Ltd (Nanjing, China). Exon5-exon6 of the *Zc3h11a* transcript was recommended as the KO region; this region contains a 244bp coding sequence. Littermates of Het-KO and WT mice were used for all experiments. Phenotypic analyses were initiated when the mice reached four weeks of age. All animals were housed in the animal husbandry room of Wenzhou Medical University. Mice were housed in standard transparent mouse cages at 22±2°C with a 12-hour light/12-hour dark cycle (light from 8 am to 8 pm, brightness approximately 200-300 lux) and free access to food and water. To exclude potential confounding effects of spontaneous ocular developmental abnormalities, a total of 7 mice (4 Het-KO and 3 WT) with small eyes or ocular lesions were excluded from the observation cohort. These anomalies were consistent with the baseline incidence of spontaneous malformations observed in historical colony data of wild-type C57BL/6J mice (approximately 11%), and were not attributed to the *Zc3h11a* heterozygous knockout. Refractive measurements were performed by a researcher blinded to the genotypes. Briefly, in a darkroom, mice were gently restrained by tail-holding on a platform facing an eccentric infrared retinoscope (EIR) (Schaeffel et al., 2004; Zhou et al., 2008a). The operator swiftly aligned the mouse position to obtain crisp Purkinje images centered on the pupil using detection software (Schaeffel et al., 2004), enabling axial measurements of refractive state and pupil size. Three repeated measurements per eye were averaged for analysis. The anterior chamber (AC) depth, lens thickness, vitreous chamber (VC)depth, and axial length (AL) of the eye were measured by real-time optical coherence tomography (a custom built OCT) (Zhou et al., 2008b). In simple terms, after anesthesia, each mouse was placed in a cylindrical holder on a positioning stage in front of the optical scanning probe. A video monitoring system was used to observe the eyes during the process. Additionally, by detecting the specular reflection on the corneal apex and the posterior lens apex in the two dimensional OCT image, the optical axis of the mouse eye was aligned with the axis of the probe. Eye dimensions were determined by moving the focal plane with a stepper motor and recording the distance between the interfaces of the eyes. Then, using the designed MATLAB software and appropriate refractive indices, the recorded optical path length was converted into geometric path length. Each eye was scanned three times, and the average value was taken. Body weight, refraction and ocular biometrics of Het-KO (n=14) and WT (n=10) mice were assessed at 4, 5, 6, 8 and 10 weeks of age.

Electroretinography

To evaluate the effect of *Zc3h11a* on the electrophysiological properties of various neuronal populations in the retina, ERG was performed on the right eye of Het-KO and WT littermates at seven weeks of age, at the same time of day (n=12). Both scotopic and photopic ERG responses were evaluated. The mice were dark-adapted overnight, and all procedures were performed under dim red light (<1 lux). The animals were anaesthetized with intraperitoneal injection of pentobarbital sodium (40 mg/kg) and the pupils were dilated with 0.5% tropicamide. A heating table (37℃) was used to maintain body temperature. ERG was recorded using a Roland Electrophysiological System (RETI-Port21, Roland Consult, Germany) with ring-shaped corneal electrodes. Scotopic ERG responses were measured under the following parameters: 2.02 log cd·s/m^2 (dark 0.01): Extremely low light intensity, assessing rod cell function. 0.48 log cd·s/m^2 (dark 3.0): Moderate light intensity, evaluating the rod-bipolar cell pathway.

0.98 log cd·s/m^2 (dark 10.0): Higher light intensity, examining bipolar cell responses under strong stimulation. For photopic ERG responses, after 10 minutes of light adaptation at 25 cd·s/m^2, recordings were performed at 0.48 log cd·s/m^2 (light 3.0). Measured parameters included: a-wave amplitude (reflecting photoreceptor function). b-wave amplitude (indicating bipolar cell activity). The responses of the Het-KO eyes (right) were compared with those of the WT eyes (right).

Immunofluorescence

Whole mouse eyes (10 weeks, littermates Het-KO and WT) were fixed with 4% para- formaldehyde (BL539A, Biosharp, China) for 1 h and 30% sucrose for dehydration overnight. Then, they were fabric-embedded in a frozen fabric matrix compound at - 20℃. Prepared tissue blocks were segmented with a cryostat at a thickness of 12 microns and collected on clean adhesive slides. The slices containing the sections were air-dried at room temperature (RT) for 15 min. After washing 3× with 1× PBS for 5 min, 5% BSA (SW3015, Solarbio, China) and 0.03% triton-X100 (P0096, Beyotime, China) diluted with 1× PBS were added as permeable membrane-blocking buffers. The slides were incubated for 1 h at RT in a humid chamber. Then, they were incubated overnight at 4℃ with the specific primary antibody Zc3h11a (1:50, 26081, Proteintech, USA). PKC α (1:200, ab32518, Abcam, UK), Opsin-1 (1:200, NB110-74730, Novus Biologicals, USA) and Rhodopsin (1:200, NBP2-25160, Novus Biologicals, USA) were added and the slices were incubated further at 4℃ in a humid chamber overnight.

After washing 4× with 1× PBS for 6 min, goat anti-rabbit Alexa Fluor 488 (1:500, ab150077, Abcam, UK) was added and the slides were incubated for 90 min at RT. After washing, the slides were mounted with an antifade medium containing DAPI (P0131, Beyotime, China) to visualize the cell nucleus. Sections incubated with 5% BSA and without primary antibodies were used as negative controls. A fluorescence microscope (LSM 880, ZEISS, Germany) was used to examine the slides and capture images. The experiments were repeated in duplicate with three different samples. Fluorescence intensity statistics were analyzed using ZEISS ZEN 3.4 software.

Transmission electron microscopy

At 10 weeks of age, two mice of each genotype were euthanized, and their eyes were removed. The eyes were fixed in a solution containing 2.5% glutaraldehyde and 0.01 M phosphate buffer (PB) (pH 7.0-7.5) for 15 min while the optic cups were dissected. The optic cups were then fixed with 1% osmium acid at room temperature away from light for 2h, after which they were rinsed three times with 0.1 MPB for 15min each time. Tissues were sequentially dehydrated in 30%-50%-70%-80%-95%-100%-100% ethanol upstream for 20 min each time, and 100% acetone twice for 15 min each time. Finally, the tissues were embedded in epoxy resin (Polybed 812) mixed 1:1 with acetone. Ultrathin sections were prepared using diamond knives and an EM UC7 ultramicrotome (Leica, Germany), then the sections were stained with 2% aqueous dioxygen acetate and 1% phosphotungstic acid (pH 3.2). Finally, the structures were examined using a transmission electron microscope (HT7700, Hitachi, Tokyo, Japan).

Cell culture, plasmid construction and transfected

HEK293T cell line (from Chinese Academy of Sciences) was cultured in DMEM medium containing 10% high-quality 10% fetal bovine serum (FBS) and 1% penicillin streptomycin dual antibody. Using restriction endonucleases HindIII and EcoRI to cleave the target gene product (wild type, *ZC3H11A*^{V138I}, *ZC3H11A*^{G43E}, *ZC3H11A*^{P154L} and *ZC3H11A*^{S747T}) and vector pcDNA3.1 (+), and purify the fragments. Then, under the action of T4 DNA ligase, the target gene was ligated with the vector pcDNA3.1 (+), and the ligated product was transformed into competent bacterial cells. Sequencing was used to identify and select the successfully constructed target gene expression plasmid vector. Finally, a sufficient amount of vector plasmids were obtained through ultrapure endotoxin extraction. Transfected with wild type, *ZC3H11A*^{V138I}, *ZC3H11A*^{G43E}, *ZC3H11A*^{P154L} and *ZC3H11A*^{S747T} overexpression plasmids using NDE3000 reagent (Western, China).

RNA sequencing analysis of molecular and pathway changes in the mouse retina

Retinas were harvested from four-week-old mice for RNA sequencing. Specifically, mice at four weeks of age were euthanized via CO_2 asphyxiation followed by cervical dislocation. Eyes were immediately enucleated and dissected to isolate intact retinas. Three retinas per group (*Zc3h11a* Het-KO and WT) were processed for transcriptomic analysis by Biomarker Biotechnology Co. (Beijing, China). The RNA sequences were mapped to the genome (GRCm38). WebGestalt (*http://www.webgestalt.org/*) was used to generate GO terms and KEGG pathways.

RNA extraction and qRT-PCR

To determine the reliability of the transcriptome results, qPCR validation was performed. Total RNA and nuclear RNA were extracted using Trizol reagent (RC112, Vazyme, China) and Cytodynamic & Nuclear RNA Purification Kit (21000, Norgen Biotek, Canada), respectively, and the purity was confirmed by the OD260/280 nm absorption ratio (1.9-2.1) (Nanodrop 2000, Thermo Scientific, USA). Total RNA (2 µg) was reverse transcribed to cDNA using a cDNA Synthesis Kit (R323, Vazyme, China). qPCR was performed with a RT-PCR detection system (Applied Biosystems, California, USA) using a SYBR Premix Ex Taq Kit (Q711, Vazyme, China), according to the manufacturer’s instructions. qRT-PCR was performed (ABI-Q6, California, USA) in a 20 µL reaction, under the following conditions: 95°C for 10 min, followed by 40 cycles of amplification at 95°C for 10 s and 60°C for 60 s. Melting curve analysis was used to determine specific amplification. All experiments were performed in triplicate. Relative quantification was performed using the ΔΔCt method. The specific gene products were amplified using the following primer pairs (**Supplement Table 1**).

Western blot

The mouse retina was separated immediately after enucleation of the eyeball and lysed in RIPA (10 mM Tris-Cl, 100 mM NaCl, 1 mM EDTA, 1 mM NaF, 20 mM $Na_4P_2O_7$, 2 mM Na_3VO_4, 1% Triton X-100, 10% glycerol, 0.1% sodium dodecyl sulphate and 0.5% deoxycholate) lysis buffer containing protease and phosphatase inhibitors (P1045, Beyotime, China). Equal amounts of protein (15 µg) were separated on 10% Tris- glycine gel, transferred to PVDF (polyvinylidene fluoride) membranes, and blocked with 5% skim milk. The primary antibodies used included ZC3H11A (ab99930, Abcam, UK), AKT (#4691, Cell Signaling Technology, USA), p-AKT (#4060, Cell Signaling Technology, USA), IκBα (ab32518, Abcam, UK), NF-κB (ab32536, Abcam, UK), TGF-β1 (21898, Proteintech, USA), PKCα (ab32518, Abcam, UK) and MMP-2 (ab92536, Abcam, UK) diluted to 1:1000 in TBST-5% milk. Then, the membranes were incubated with goat anti-rabbit IgG conjugated with HRP (1:2000 in TBST-5% milk) (SA00001-2, Proteintech, USA) for 90 min at room temperature and developed using western blotting reagents (BL523B, Bioshark, China). GAPDH (AF2823, Byotime, China) was used as the internal control.

Software and statistical analysis

All experiments were repeated at least once, and sample sizes and reported results reflect the cumulative data for all trials of each experiment. Each result is expressed as the mean ± standard deviation (SD). Pearson’s test was used to assess the normality of the data. The normally distributed data were subjected to parametric analyses. Unpaired Student’s t-tests were used for parametric analyses between two groups. Data that were not normally distributed or had a sample size that was too small, were subjected to nonparametric analyses. Nonparametric tests between two groups were performed using the Wilcoxon signed-rank test for matched pairs, the Mann-Whitney U-test or the Kruskal-Wallis test for multiple comparisons (GraphPad Prism 9, La Jolla, CA). A difference was considered statistically significant when the p-value was less than 0.05 and highly significant if it was less than 0.01 or less than 0.001.

Data availability

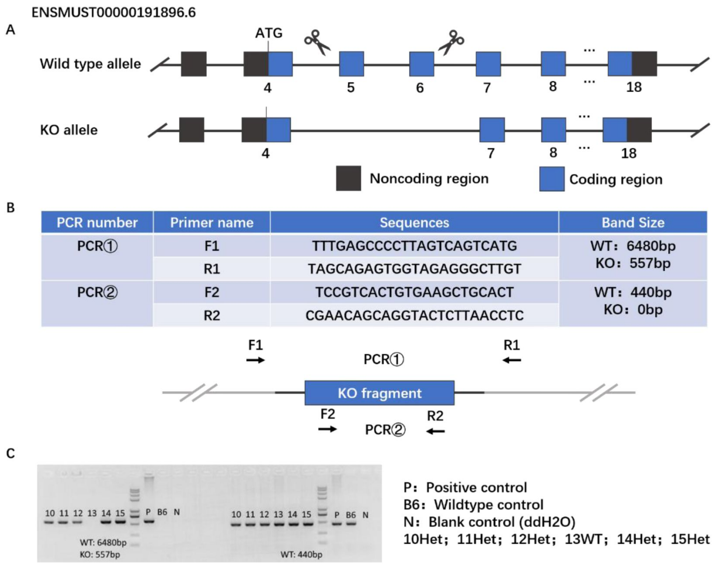
All data generated or analysed during this study are included in the manuscript and supporting files.

Supplementary files

Supplement Figure 1.

Generation of *Zc3h11a* KO mice.

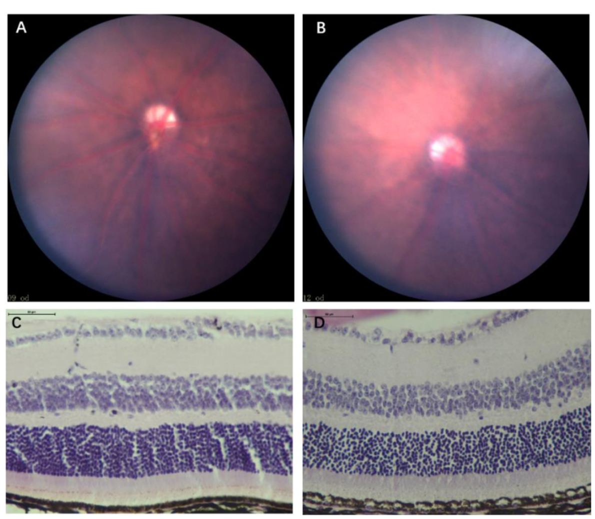
(A) Generation of Zc3h11a KO mice in C57BL/6J background using CRISPR/Cas9 technology: Exon 5 to exon 6 of the *Zc3h11a* transcript was used as the KO region; this region contains a 244 bp coding sequence, and knocking out this region will result in loss of protein function (B-C) Primers for genotyping and examples of genotyping results of *Zc3h11a* Het-KO mice and wild-type mice.



Supplement Figure 2.

Fundus photographs and HE staining of Het and WT mice at 8th week.

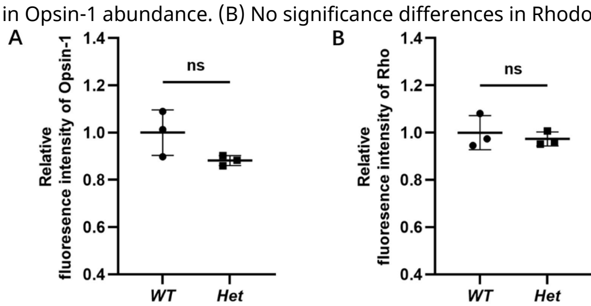
(A. B) There were no significant differences in fundus photographs between *Zc3h11a* Het-KO and WT mice, WT (A) and Het-KO mice (B) (n=3). (C. D) There were no significant differences in HE staining between *Zc3h11a* Het-KO and WT mice. WT (C) and Het-KO mice (D) (n=3).



Supplement Figure 3.

Quantitative analysis of cone cells and rod cells in Zc3h11a Het-KO.

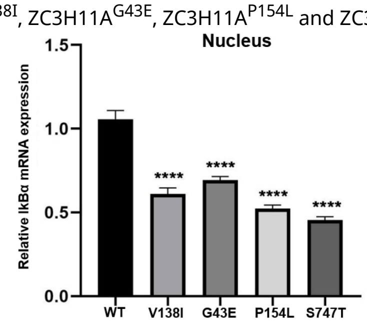
(A) No significance differences in Opsin-1 abundance. (B) No significance differences in Rhodopsin protein abundance.



Supplement Figure 4.

The relative mRNA expression levels of IκBα in the nucleus.

Transfection of ZC3H11A overexpression mutant plasmid into cells resulted in a significant decrease in the mRNA expression levels of IκBα in four mutants (ZC3H11A^{V138I}, ZC3H11A^{G43E}, ZC3H11A^{P154L} and ZC3H11A^{S747T}).



Supplement Table 1.

Sequence of oligonucleotides

Gene	Sequence
ZC3H11A	Forward: CTAAACTGCGCTTTCCATCACA Reserve: CGTTGATTACAACAGGCGGAT
PI3K	Forward: TGGGACCTTTTTGGTACGAGA Reserve: AGCTAAAGACTCATTCCGGTAGT
AKT	Forward: CCTTTATTGGCTACAAGGAACGG Reserve: GAAGGTGCGCTCAATGACTG
IκBα	Forward: TGAAGGACGAGGAGTACGAGC Reserve: TGCAGGAACGAGTCTCCGT
NF-κB	Forward: GGGGCCTGCAAAGGTTATC Reserve: TGCTGTTACGGTGCATACCC
TGF-β1	Forward: GTAACGCCAGCCAGGAATTGTTGCTA Reserve: CTTCAATACGTCAGACATTCGGG
IL-6	Forward: CTGCAAGAGACTTCCATCCAG Reserve: AGTGGTATAGACAGGTCTGTTGG
MMP-2	Forward: CAAGTTCCCCGGCGATGTC Reserve: TTCTGGTCAAGGTCACCTGTC
GAPDH	Forward: AGGTCGGTGTGAACGGCTTTG Reserve: TGTAGACCATCTAGTTGAGGTCA

The mRNA expression levels were analysed by the comparative $2^{-\Delta\Delta}$ Ct method and normalised using GAPDH.

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Author Contributions

Conceptualisation, Xinting Liu, Jia Qu and Fan Lu; methodology, Chong Chen, Qian Liu and Cheng Tang; data analysis, Chong Chen, Qian Liu, Cheng Tang and Yu Rong; investigation and cohort construction, Xinting Liu, Jia Qu, Fan Lu, Dandan Li and Xinyi Zhao; writing original draft preparation, Chong Chen, Qian Liu and Xinting Liu; writing-review and editing, Xinting Liu, Jia Qu and Fan Lu; funding acquisition, Xinting Liu, Jia Qu and Fan Lu. All authors have read and agreed to the published version of the manuscript.

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References

Alzahrani A.S (2019) **PI3K/Akt/mTOR inhibitors in cancer: At the bench and bedside** *Semin Cancer Biol* **59**:125–132 Google Scholar

Baird P.N., Saw S.M., Lanca C., Guggenheim J.A., Smith Iii E.L., Zhou X., Matsui K.O., Wu P.C., Sankaridurg P., Chia A., Rosman M., Lamoureux E.L., Man R., He M (2020) **Myopia** *Nat Rev Dis Primers* **6**:99 Google Scholar

Baron V., Adamson E.D., Calogero A., Ragona G., Mercola D (2006) **The transcription factor Egr1 is a direct regulator of multiple tumor suppressors including TGFbeta1, PTEN, p53, and fibronectin** *Cancer Gene Ther* Google Scholar

Cai X.-B., Zheng Y.-H., Chen D.-F., Zhou F.-Y., Xia L.-Q., Wen X.-R., Yuan Y.-M., Han F., Piao S.-Y., Zhuang W., Lu F., Qu J., Yu A.Y., Jin Z.-B (2019) **Expanding the Phenotypic and Genotypic Landscape of Nonsyndromic High Myopia: A Cross-Sectional Study in 731 Chinese Patients** *Invest Ophthalmol Vis Sci* **60**:4052–4062 Google Scholar

Chen B.-Y., Wang C.-Y., Chen W.-Y., Ma J.-X (2013) **Altered TGF-β2 and bFGF expression in scleral desmocytes from an experimentally-induced myopia guinea pig model** *Graefes Arch Clin Exp Ophthalmol* **251**:1133–1144 Google Scholar

Chen C.-S., Hsu Y.-A., Lin C.-H., Wang Y.-C., Lin E.-S., Chang C.-Y., Chen J.J.-Y., Wu M.-Y., Lin H.-J., Wan L. (2022) **Fallopia Japonica and Prunella vulgaris inhibit myopia progression by suppressing AKT and NFκB mediated inflammatory reactions** *BMC Complement Med Ther* Google Scholar

Darweesh M., Younis S., Hajikhezri Z., Ali A., Jin C., Punga T., Gupta S., Essand M., Andersson L., Akusjärvi G (2022) **ZC3H11A loss of function enhances NF-κB signaling through defective IκBα protein expression** *Frontiers in Immunology* **13**:1002823 Google Scholar

Dyson H.J., Komives E.A (2012) **Role of disorder in IκB-NFκB interaction** *IUBMB Life* **64**:499–505 Google Scholar

Fan Q., Barathi V.A., Cheng C.-Y., Zhou X., Meguro A., Nakata I., Khor C.-C., Goh L.-K., Li Y.-J., Lim W.e., Ho C.E.H., Hawthorne F., Zheng Y., Chua D., Inoko H., Yamashiro K., Ohno-Matsui K., Matsuo K., Matsuda F., Vithana E., Seielstad M., Mizuki N., Beuerman R.W., Tai E.S., Yoshimura N., Aung T., Young T.L., Wong T.-Y., Teo Y.-Y., Saw S.-M. (2012) **Genetic variants on chromosome 1q41 influence ocular axial length and high myopia** *PLoS Genet* **8**:e1002753 Google Scholar

Flitcroft D.I., He M., Jonas J.B., Jong M., Naidoo K., Ohno-Matsui K., Rahi J., Resnikoff S., Vitale S., Yannuzzi L (2019) **IMI - Defining and Classifying Myopia: A Proposed Set of Standards for Clinical and Epidemiologic Studies** *Invest Ophthalmol Vis Sci* **60**:M20–M30 Google Scholar

Haarman A.E.G., Thiadens A.A.H.J., van Tienhoven M., Loudon S.E., de Klein J.E.M.M.A., Brosens E., Polling J.R., van der Schoot V., Bouman A., Kievit A.J.A., Hoefsloot L.H., Klaver C.C.W., Verhoeven V.J.M. (2022) **Whole exome sequencing of known eye genes reveals genetic causes for high myopia** *Hum Mol Genet* **31**:3290–3298 Google Scholar

Harper A.R., Summers J.A (2015) **The dynamic sclera: extracellular matrix remodeling in normal ocular growth and myopia development** *Exp Eye Res* **133**:100–111 Google Scholar

Hayden M.S., Ghosh S (2011) **NF-κB in immunobiology** *Cell Res* **21**:223–244 Google Scholar

Holden B.A., Fricke T.R., Wilson D.A., Jong M., Naidoo K.S., Sankaridurg P., Wong T.Y., Naduvilath T.J., Resnikoff S (2016) **Global Prevalence of Myopia and High Myopia and Temporal Trends from 2000 through 2050** *Ophthalmology* **123**:1036–1042 Google Scholar

Ikeda S.-I., Kurihara T., Jiang X., Miwa Y., Lee D., Serizawa N., Jeong H., Mori K., Katada Y., Kunimi H., Ozawa N., Shoda C., Ibuki M., Negishi K., Torii H., Tsubota K (2022) **Scleral PERK and ATF6 as targets of myopic axial elongation of mouse eyes** *Nat Commun* **13**:5859 Google Scholar

Jimi E., Fei H., Nakatomi C (2019) **NF-κB Signaling Regulates Physiological and Pathological Chondrogenesis** *Int J Mol Sci* **20** Google Scholar

Koga M., Mizuno Y., Watanabe I., Kawakami H., Goto T (2014) **Role of VPAC2 receptor in monocrotaline-induced pulmonary hypertension in rats** *J Appl Physiol* Google Scholar

Krizaj D., Copenhagen D.R (2002) **Calcium regulation in photoreceptors** *Frontiers in bioscience: a journal and virtual library* **7**:d2023 Google Scholar

Libermann T.A., Baltimore D (1990) **Activation of interleukin-6 gene expression through the NF-kappa B transcription factor** *Mol Cell Biol* **10**:2327–2334 Google Scholar

Lin H.-J., Wei C.-C., Chang C.-Y., Chen T.-H., Hsu Y.-A., Hsieh Y.-C., Chen H.-J., Wan L (2016) **Role of chronic inflammation in myopia progression: clinical evidence and experimental validation** *EBioMedicine* **10**:269–281 Google Scholar

Liu S., Chen T., Chen B., Liu Y., Lu X., Li J (2022) **Lrpap1 deficiency leads to myopia through TGF-β-induced apoptosis in zebrafish** *Cell Commun Signal* **20**:162 Google Scholar

Manavalan B., Basith S., Choi Y.-M., Lee G., Choi S (2010) **Structure-function relationship of cytoplasmic and nuclear IκB proteins: an in silico analysis** *PLoS One* **5**:e15782 Google Scholar

Morgan M.J., Liu Z.-g. (2011) **Crosstalk of reactive oxygen species and NF-κB signaling** *Cell Res* **21**:103–115 Google Scholar

Qi J., Pan W., Peng T., Zeng L., Li X., Chen Z., Yang Z., Xu H (2022) **Higher Circulating Levels of Neutrophils and Basophils Are Linked to Myopic Retinopathy** *Int J Mol Sci* **24** Google Scholar

Saw S.-M., Gazzard G., Shih-Yen E.C., Chua W.-H (2005) **Myopia and associated pathological complications** *Ophthalmic Physiol Opt* **25**:381–391 Google Scholar

Schaeffel F., Burkhardt E., Howland H.C., Williams R.W (2004) **Measurement of refractive state and deprivation myopia in two strains of mice** *Optometry and Vision Science* **81**:99–110 Google Scholar

Schippert R., Burkhardt E., Feldkaemper M., Schaeffel F (2007) **Relative axial myopia in Egr-1 (ZENK) knockout mice** *Invest Ophthalmol Vis Sci* **48**:11–17 Google Scholar

Shi Y., Li Y., Zhang D., Zhang H., Li Y., Lu F., Liu X., He F., Gong B., Cai L., Li R., Liao S., Ma S., Lin H., Cheng J., Zheng H., Shan Y., Chen B., Hu J., Jin X., Zhao P., Chen Y., Zhang Y., Lin Y., Li X., Fan Y., Yang H., Wang J., Yang Z (2011) **Exome sequencing identifies ZNF644 mutations in high myopia** *PLoS Genet* **7**:e1002084 Google Scholar

Szczerkowska K.I., Petrezselyova S., Lindovsky J., Palkova M., Dvorak J., Makovicky P., Fang M., Jiang C., Chen L., Shi M., Liu X., Zhang J., Kubik-Zahorodna A., Schuster B., Beck I.M., Novosadova V., Prochazka J., Sedlacek R (2019) **Myopia disease mouse models: a missense point mutation (S673G) and a protein-truncating mutation of the Zfp644 mimic human disease phenotype** *Cell Biosci* **9**:21 Google Scholar

Tang S.M., Li F.F., Lu S.Y., Kam K.W., Tam P.O.S., Tham C.C., Pang C.P., Yam J.C.S., Chen L.J (2020) **Association of the ZC3H11B, ZFHX1B and SNTB1 genes with myopia of different severities** *Br J Ophthalmol* **104**:1472–1476 Google Scholar

Tedja M.S., Haarman A.E.G., Meester-Smoor M.A., Kaprio J., Mackey D.A., Guggenheim J.A., Hammond C.J., Verhoeven V.J.M., Klaver C.C.W. (2019) **IMI - Myopia Genetics Report** *Invest Ophthalmol Vis Sci* **60** Google Scholar

Tian Q., Tong P., Chen G., Deng M., Cai T.e., Tian R., Zhang Z., Xia K., Hu Z. (2023) **GLRA2 gene mutations cause high myopia in humans and mice** *J Med Genet* **60**:193–203 Google Scholar

Tkatchenko A.V., Walsh P.A., Tkatchenko T.V., Gustincich S., Raviola E (2006) **Form deprivation modulates retinal neurogenesis in primate experimental myopia** *Proc Natl Acad Sci U S A* **103**:4681–4686 Google Scholar

Tkatchenko T.V., Tkatchenko A.V (2019) **Pharmacogenomic Approach to Antimyopia Drug Development: Pathways Lead the Way** *Trends Pharmacol Sci* **40**:833–852 Google Scholar

Tkatchenko T.V., Troilo D., Benavente-Perez A., Tkatchenko A.V (2018) **Gene expression in response to optical defocus of opposite signs reveals bidirectional mechanism of visually guided eye growth** *PLoS Biol* **16**:e2006021 Google Scholar

Torrealba N., Vera R., Fraile B., Martínez-Onsurbe P., Paniagua R., Royuela M (2020) **TGF-β/PI3K/AKT/mTOR/NF-kB pathway. Clinicopathological features in prostate cancer** *Aging Male* **23**:801–811 Google Scholar

Ugarte M., Osborne N.N (2001) **Zinc in the retina** *Progress in neurobiology* **64**:219–249 Google Scholar

Wallman J., Winawer J (2004) **Homeostasis of eye growth and the question of myopia** *Neuron* **43**:447–468 Google Scholar

Wang G.-B., Zhang Q.-J., Xiao X.-S., Li J.-Z., Zhang F.-S., Li S.-Q., Li W., Li T., Jia X.-Y., Guo L., Guo X.-M (2004) **[Variations of the zinc finger protein 161 gene in Chinese with or without high myopia]** *Yi Chuan* **26**:155–159 Google Scholar

Wu K.-I.S., Schmid-Schönbein G.W (2011) **Nuclear factor kappa B and matrix metalloproteinase induced receptor cleavage in the spontaneously hypertensive rat** *Hypertension* **57**:261–268 Google Scholar

Xiao K., Chen Z., He S., Long Q., Chen Y (2025) **The role of NF-κB pathway and its regulation of inflammatory cytokines in scleral remodeling of form-deprivation mice model** *Immunologic Research* **73**:1–8 Google Scholar

Xiao Q., Xiao J., Liu J., Liu J., Shu G., Yin G (2022) **Metformin suppresses the growth of colorectal cancer by targeting INHBA to inhibit TGF-β/PI3K/AKT signaling transduction** *Cell Death Dis* **13**:202 Google Scholar

Yang E., Yu J., Liu X., Chu H., Li L (2023) **Familial Whole Exome Sequencing Study of 30 Families With Early-Onset High Myopia** *Invest Ophthalmol Vis Sci* **64**:10 Google Scholar

Ye M., Ma Y., Qin Y.-X., Cai B., Ma L.-M., Ma Z., Liu Y., Jin Z.-B., Zhuang W.-J (2023) **Mutational investigation of 17 causative genes in a cohort of 113 families with nonsyndromic early-onset high myopia in northwestern China** *Mol Genet Genomics* **298**:669–682 Google Scholar

Yoshida M., Whitsett J.A (2006) **Alveolar macrophages and emphysema in surfactant protein-D-deficient mice** *Respirology* **11**:S37–S40 Google Scholar

Younis S., Jouneau A., Larsson M., Oudin J.F., Adenot P., Omar J., Brochard V., Andersson L (2023) **Ablation of ZC3H11A causes early embryonic lethality and dysregulation of metabolic processes** *Proc Natl Acad Sci U S A* **120**:e2216799120 Google Scholar

Younis S., Kamel W., Falkeborn T., Wang H., Yu D., Daniels R., Essand M., Hinkula J., Akusjärvi G., Andersson L (2018) **Multiple nuclear-replicating viruses require the stress-induced protein ZC3H11A for efficient growth** *Proc Natl Acad Sci U S A* **115**:E3808–E3816 Google Scholar

Zhao F., Zhou Q., Reinach P.S., Yang J., Ma L., Wang X., Wen Y., Srinivasalu N., Qu J., Zhou X (2018) **Cause and Effect Relationship between Changes in Scleral Matrix Metallopeptidase-2 Expression and Myopia Development in Mice** *Am J Pathol* **188**:1754–1767 Google Scholar

Zhou X., Shen M., Xie J., Wang J., Jiang L., Pan M., Qu J., Lu F (2008a) **The development of the refractive status and ocular growth in C57BL/6 mice** *Investigative ophthalmology & visual science* **49**:5208–5214 Google Scholar

Zhou X., Xie J., Shen M., Wang J., Jiang L., Qu J., Lu F (2008b) **Biometric measurement of the mouse eye using optical coherence tomography with focal plane advancement** *Vision research* **48**:1137–1143 Google Scholar

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Reviewer #1 (Public Review):

The authors reported that mutations were identified in the ZC3H11A gene in four adolescents from 1015 high myopia subjects in their myopia cohort. They further generated Zc3h11a knockout mice utilizing the CRISPR/Cas9 technology.

The main claims are only partially supported. The reviewers still have the concerns of 1) the modes of inheritance for the families need to be shown; 2) the phenotype of heterozygous mutant mice is too weak; 3) the authors still have not addressed the biological question of whether there are fewer bipolar cells or decreased expression of the marker protein. This would involve counting cells, which they have not done. The blots they show do not appear to support their quantifications. Considering the sensitivity of quantifying nearly saturated blots, the authors should show blots that are not exposed to that level of saturation.

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Author response:

The following is the authors' response to the previous reviews.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *The authors reported that mutations were identified in the ZC3H11A gene in four adolescents from 1015 high myopia subjects in their myopia cohort. They further generated Zc3h11a knockout mice utilizing the CRISPR/Cas9 technology.*
>
> *Comments on revisions:*
>
> *Chong Chen and colleagues revised the manuscript; however, none of my suggestions from the initial review have been sufficiently addressed.*
>
> *(1) I indicated that the pathogenicity and novelty of the mutation need to be determined according to established guidelines and databases. However, the conclusion was still drawn without sufficient justification.*

Thank you for your valuable feedback on the assessment of mutation pathogenicity and novelty. We regret to inform you that complete familial genetic information required for segregation analysis is currently unavailable in this study. Despite our exhaustive efforts to contact the four mutation carriers and their relatives, we encountered the following uncontrollable limitations: Two patients could not be further traced due to invalid contact information, one patient had relocated to another region, making sample collection logistically unfeasible, the remaining patient explicitly declined family participation in genetic testing due to privacy concerns.

We fully acknowledge that the lack of pedigree data may affect the certainty of pathogenicity evaluation. To address this limitation, we systematically analyzed the four *ZC3H11A* missense mutations (c.412G>A p.V138I, c.128G>A p.G43E, c.461C>T p.P154L, and c.2239T>A p.S747T) based on ACMG guidelines and database evidence. The key findings are summarized below: All of the identified mutations exhibited very low frequencies or does not exist in the Genome Aggregation Database (gnomAD) and Clinvar, and using pathogenicity prediction software SIFT, PolyPhen2, and CADD, most of them display high pathogenicity levels. Among them, c.412G>A, c.128G>A and c.461C>T were located in or around a domain named zf-CCCH_3 (Figure 1A and B). Furthermore, all of the mutation sites were located in highly conserved amino acids across different species (Figure 1C). The four mutations induced higher structural flexibility and altered the negative charge at corresponding sites, potentially disrupting protein-RNA interactions (Figure 1D and E). Concurrently, overexpression of mutant constructs (*ZC3H11A*-V138I, *ZC3H11A*-G43E, *ZC3H11A*-P154L, and *ZC3H11A*-S747T) revealed significantly reduced nuclear IκBα mRNA levels compared to the wild-type, suggesting impaired NF-κB pathway regulation (Supplementary Figure 4). Zc3h11a knockout mice also exhibited a myopic phenotype, with alterations in the PI3K-AKT and NF-κB signaling pathways. Integrating this evidence, the mutations meet the following ACMG criteria: PM1 (domain-located mutations), PM2 (extremely low population frequency), PP3 (computational predictions supporting pathogenicity), PS3 (functional validation via experimental assays). Under the ACMG framework, these mutations are classified as "Likely Pathogenic".

Regarding the novelty of this mutation, comprehensive searches in ClinVar, dbSNP, and HGMD databases revealed no prior reports associating this variant with myopia. Similarly, a PubMed literature search identified no direct evidence linking this mutation to myopia. Based on this evidence, we classify this variant as a likely pathogenic and novel mutation.

On the other hand, we acknowledge that the absence of family segregation data may reduce the confidence in pathogenicity assessment. Nevertheless, functional experiments and converging multi-level evidence strongly support the reliability of our conclusion. Future studies will prioritize family-based validation to strengthen the evidence chain. We sincerely appreciate your attention to this matter and kindly request your understanding of the practical limitations inherent to this research.

> *(2) The phenotype of heterozygous mutant mice is too weak to support the gene's contribution to high myopia. The revised manuscript does not adequately address these discrepancies. Furthermore, no explanation was provided for why conditional gene deletion was not used to avoid embryonic lethality, nor was there any discussion on tissue- or cell-specific mechanistic investigations.*

We sincerely appreciate your insightful comments regarding the relationship between murine phenotypes and human disease. We fully acknowledge your concerns about the phenotypic strength of *Zc3h11a* heterozygous mutant mice and their association with high myopia (HM) pathogenesis. Here we provide point-by-point responses to your valuable comments: Our study demonstrates that Zc3h11a heterozygous mutant mice exhibit myopic refractive phenotypes with upregulated myopia-associated factors (TGF-β1, MMP2, and IL6), although axial elongation did not reach statistical significance. Notably, at 4 and 6 weeks of age, Het mice did display longer axial lengths and vitreous chamber depths compared to WT mice. While these differences did not reach statistical significance at other time points, an increasing trend was still observed. Several technical considerations may explain these findings: The small murine eye size (where 1D refractive change corresponds to only 5-6μm axial length change). The theoretical resolution limit of 6μm for the SD-OCT device used in this study. These factors likely contributed to the marginal statistical significance observed in the subtle changes of vitreous chamber depth and axial length measurements. Additionally, existing research indicates that axial length measurements from frozen sections in age-matched mice tend to be longer than those obtained through in vivo measurements. This phenomenon may reflect species differences between humans and mice - while both show significant refractive power changes, the axial length differences are less pronounced in mice. These results align with previous reports of phenotypic differences between mouse models and human myopia.

To address these issues comprehensively, we have added a dedicated discussion section in the revised manuscript specifically examining these axial length measurement considerations, following your valuable suggestion.

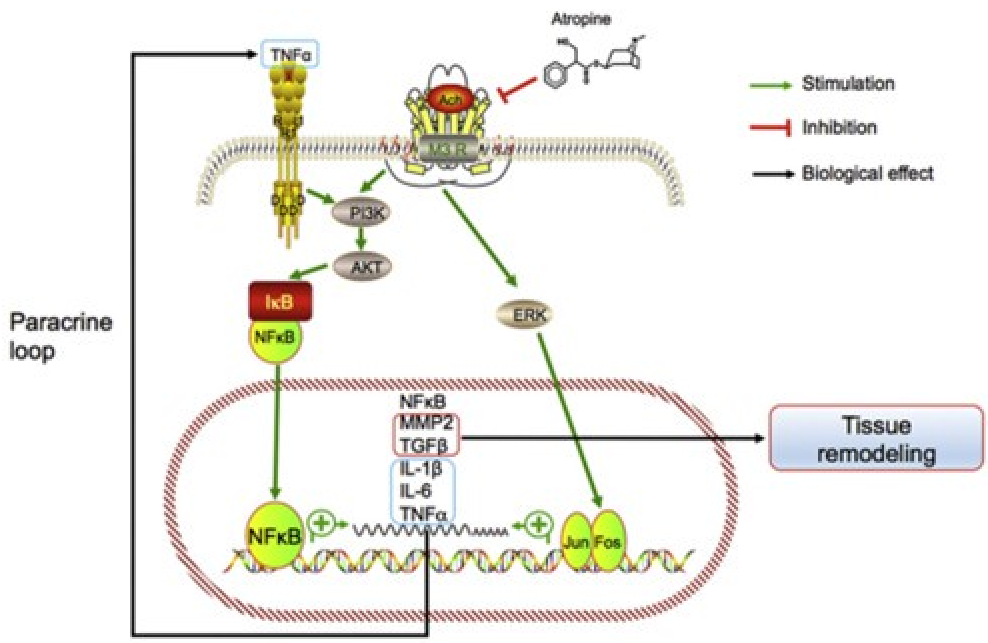
Additionally, we regret to inform you that the currently available floxed ZC3H11A mouse strain requires a minimum of 12-18 months for custom construction, which exceeds our research timeline due to current resource limitations in our team. To address this gap, we have supplemented the discussion section with additional content regarding tissue- and cell-specific mechanisms. Based on your constructive suggestions, we will prioritize the following in our subsequent work: Collaborate with transgenic animal centers to generate *Zc3h11a* conditional knockout mice. Evaluate the impact of specific knockouts on myopia progression using form-deprivation (FDM) models. While we recognize the limitations of our current study, we believe that by integrating clinical cohort data, phenotypic evidence, and functional experiments, this research provides valuable directional evidence for *ZC3H11A*'s potential role in myopia pathogenesis. Your comments will significantly contribute to improving our future research design, and we sincerely hope you can recognize the exploratory significance of our current findings.

> *(3) The title, abstract, and main text continue to misrepresent the role of the inflammatory intracellular PI3K-AKT and NF-κB signaling cascade in inducing high*

> *myopia. No specific cell types have been identified as contributors to the phenotype. The mice did not develop high myopia, and no relationship between intracellular signaling and myopia progression has been demonstrated in this study.*

Thank you for your valuable comments regarding the interpretation of signaling pathways in our study. We fully acknowledge your rigorous concerns about the role of PI3K-AKT and NF-κB signaling cascades in high myopia and recognize that we did not identify specific cell types contributing to the observed phenotype. In response to your feedback, we have removed the hypothetical statement linking genetic changes within inflammatory cells to the development of myopia. The current interpretation is strictly based on experimental evidence of pathway relevance and is supported by the theoretical basis presented in the reference, specifically that loss of Zc3h11a leads to activation of the PI3K-AKT and NF-κB pathways in retinal cells, contributing to the myopic phenotype.

Author response image 1.



Model of the association between inflammation and myopia progression. Activated mAChR3 (M3R) activates phosphoinositide 3-kinase (PI3K)–AKT and mitogen-associated protein kinase (MAPK) signaling pathways, in turn activating NF-κB and AP1 (i.e., the Jun.-Fos heterodimer) and stimulating the expression of the target genes NF-κB, MMP2, TGFβ, IL- 1β and -6, and TNF-α. MMP2 and TGF-β promote tissue remodeling and TNF-α may act in a paracrine feedback loop in the retina or sclera to activate NF-κB during myopia progression.

To address the limitations raised, we will prioritize the following in future studies: Cell-type-specific knockout models to identify key cellular contributors. Mechanistic investigations to establish causal relationships between signaling pathways and myopia progression. We sincerely appreciate your rigorous review, which has significantly improved the scientific accuracy and clarity of our manuscript. We believe the revised version better reflects both the novelty and limitations of our findings. We kindly request your recognition of the study's contributions while acknowledging its current constraints.

Reviewer #3 (Public review):

Chen et al have identified a new candidate gene for high myopia, ZC3H11A, and using a knock-out mouse model, have attempted to validate it as a myopia gene and explain a potential mechanism. They identified 4 heterozygous missense variants in highly myopic teenagers. These variants are in conserved regions of the protein, and predicted to be damaging, but the only evidence the authors provide that these specific variants affect protein function is a supplement figure showing decreased levels of IκBα after transfection with overexpression plasmids (not specified what type of cells were transfected). This does not prove that these mutations cause loss of function, in fact it implies they have a gain-of-function mechanism. They then created a knock-out mouse. Heterozygotes show myopia at all ages examined but increased axial length only at very early ages. Unfortunately, the authors do not address this point or examine corneal structure in these animals. They show that the mice have decreased B-wave amplitude on electroretinogram (a sign of retinal dysfunction associated with bipolar cells), and decreased expression of a bipolar cell marker, PKCα. On electron microscopy, there are morphologic differences in the outer nuclear layer (where bipolar, amacrine, and horizontal cell bodies reside). Transcriptome analysis identified over 700 differentially expressed genes. The authors chose to focus on the PI3K-AKT and NF-κB signaling pathways and show changes in expression of genes and proteins in those pathways, including PI3K, AKT, IκBα, NF-κB, TGF-β1, MMP-2 and IL-6, although there is very high variability between animals. They propose that myopia may develop in these animals either as a result of visual abnormality (decreased bipolar cell function in the retina) or by alteration of NF-κB signaling. These data provide an interesting new candidate variant for development of high myopia, and provide additional data that MMP2 and IL6 have a role in myopia development. For this revision, none of my previous suggestions have been addressed.

Reviewer #3 (Recommendations for the authors):

None of these suggestions were addressed in the revision:

Major issues:

(1) Figure 2: refraction is more myopic but axial length is not longer - why is this not discussed and explored? The text claims the axial length is longer, but that is not supported by the figure. If this is a measurement issue, that needs to be discussed in the text.

We sincerely appreciate your valuable comments regarding the relationship between refractive status and axial length in our study. In response to your concerns, we have conducted an in-depth analysis and would like to address the issues as follows:

Our data demonstrate significant differences in refractive error between heterozygous (Het) and wild-type (WT) mice during the 4-10 weeks. Notably, at 4 and 6 weeks of age, Het mice did exhibit longer axial lengths and greater vitreous chamber depth compared to WT mice, although these differences did not reach statistical significance at other time points while still showing an increasing trend. Additional measurements of corneal curvature revealed no significant differences between groups. Considering the small size of mouse eyes (where a 1D refractive change corresponds to only 5-6μm axial length change) and the theoretical resolution limit of 6μm for the SD-OCT device used in this study, these technical factors may account for the marginal statistical significance of the observed small changes in vitreous chamber depth and axial length measurements. Furthermore, existing studies have shown that axial length measurements from frozen sections tend to be longer than those obtained from in vivo measurements in age-matched mice. These considerations provide plausible explanations for the apparent discrepancy between refractive changes and axial length parameters. Following your suggestion, we have added a dedicated discussion section addressing these axial length measurement issues in the revised manuscript. We fully understand your concerns regarding data consistency, and your comments have prompted us to conduct more comprehensive and thorough analysis of our results. We believe the revised manuscript now more accurately reflects our findings while providing important technical references for future studies.

> *(2) Slipped into the methods is a statement that mice with small eyes or ocular lesions were excluded. How many mice were excluded? Are the authors ignoring another phenotype of these mice?*

We appreciate your attention to the exclusion criteria and their implications. Below we provide a detailed clarification: A total of 7 mice (4 Het-KO and 3 WT) with small eyes or ocular lesions were excluded from the observation cohort. These anomalies were consistent with the baseline incidence of spontaneous malformations observed in historical colony data of wild-type C57BL/6J mice (approximately 11%), and were not attributed to the *Zc3h11a* heterozygous knockout. We have added the above content in the methods section. Your insightful comment has significantly strengthened our reporting rigor. We hope this clarification alleviates your concerns regarding potential selection bias or overlooked phenotypes.

> *Minor/Word choice issues:*
>
> *All the figure legends need to be improved so that each figure can be interpreted without having to refer to the text.*

Thank you for your valuable comments. We have made modifications to the legend of each graphic, as detailed in the main text.

> *Abstract: line 24: use refraction, not "vision"*

Thank you for your valuable comments. The “Vision” has been changed to “refraction”.

> *Line 28: re-word "density of bipolar cell-labeled proteins" Do the authors mean density of bipolar cells? Or certain proteins were less abundant in bipolar cells?*

Thank you for your rigorous review of this terminology. We acknowledge the need to clarify the precise meaning of the phrase "density of bipolar cell-labeled proteins." In the original text, this term specifically refers to the expression abundance of the bipolar cell-specific marker protein PKCα, which was identified using immunofluorescence labeling techniques. Specifically: We utilized PKCα (a bipolar cell marker) to label bipolar cell populations. The "density" was quantified by measuring the fluorescence signal intensity per unit area in confocal microscopy images, rather than direct cell counting. This metric reflects changes in the expression of the specific marker protein (PKCα) within bipolar cells, which indirectly correlates with alterations in bipolar cell populations. To address ambiguity, we have revised the terminology throughout the manuscript to "bipolar cell-labelled protein PKCα immunofluorescence abundance".

Additionally, since fluorescence intensity quantification is inherently semi-quantitative, we have included Western blot results for PKCα in the revised manuscript (Figure 3I, J) to validate the expression changes observed via immunofluorescence. We sincerely appreciate your feedback, which has significantly improved the precision of our manuscript.

> *Line 45: axial length, not ocular axis*

Thank you for your valuable comments. The “ocular axis” has been changed to “axial length”.

> *Lines73-75: confusing*

Thank you for your valuable comments. The relevant content has been modified to “Multiple zinc finger protein genes (e.g., *ZNF644, ZC3H11B, ZFP161, ZENK*) are associated with myopia or HM. Of these, *ZC3H11B* (a human homolog of *ZC3H11A*) and five GWAS loci (Schippert et al., 2007; Shi et al., 2011; Szczerkowska et al., 2019; Tang et al., 2020; Wang et al., 2004) correlate with AL elongation or HM severity. Proteomic studies further suggest ZC3H11A involvement in the TREX complex, implicating RNA export mechanisms in myopia pathogenesis”

> *Line 138: what is dark 3.0 and dark 10.0*

Thank you for your valuable comments. The relevant content has been modified to “Upon dark adaptation, b-wave amplitudes in seven-week-old Het-KO mice were significantly lower at dark 3.0 (0.48 log cd·s/m^2) and dark 10.0 (0.98 log cd·s/m^2) compared to WT mice.” A detailed description has been added to the main text methods.

> *Line 171-175: the GO terms of "biological processes" and "molecular functions" are so broad as to be meaningless.*

Thank you for your valuable comments. The relevant content has been modified to “GO enrichment analysis revealed significant enrichment of differentially expressed genes in the following functions: Zinc ion transmembrane transport (GO:0071577) within metal ion homeostasis, associated with retinal photoreceptor maintenance (Ugarte and Osborne, 2001), RNA biosynthesis and metabolism (GO:0006366) in transcriptional regulation, potentially influencing ocular development, negative regulation of NF-κB signaling (GO:0043124) in inflammatory modulation, a pathway involved in scleral remodelling (Xiao et al., 2025), calcium ion binding (GO:0005509), critical for phototransduction (Krizaj and Copenhagen, 2002), zinc ion transmembrane transporter activity (GO:0005385), participating in retinal zinc homeostasis (Figure 5C and D).”

> *Line 257-259: which results indicated loss of Zc3h11a inhibited translocation of IκBα from nucleus to cytoplasm? Results of this study, or the previously referenced study?*

We sincerely appreciate your critical inquiry regarding the mechanistic relationship between *Zc3h11a* deficiency and IκBα translocation. We are grateful for this opportunity to clarify this important point. The findings regarding *Zc3h11a*-mediated regulation of IκBα mRNA nuclear export and its impact on NF-κB signaling originate from the study by Darweesh et al. The key experimental evidence demonstrates that: The depletion of *Zc3h11a* leads to nuclear retention of IκBα mRNA, resulting in failure to maintain normal levels of cytoplasmic IκBα mRNA and protein. This defect in IκBα mRNA export disrupts the essential inhibitory feedback loop on NF-κB activity, causing hyperactivation of this pathway. This manifests as upregulation of numerous innate immune-related mRNAs, including IL-6 and a large group of interferon-stimulated genes.While our study references this mechanism to explain the observed NF-κB dysregulation in *Zc3h11a* Het-KO mice, the specific nuclear export mechanism was indeed elucidated by Darweesh et al. The reference has been inserted into the corresponding position in the main text. Importantly, our research extends these previous molecular insights into the phenotypic context of myopia.

We sincerely regret any ambiguity in the original text and deeply appreciate your rigorous approach in ensuring proper attribution of these fundamental findings. Your comment has significantly improved the clarity and accuracy of our manuscript.

> *Figure 6 shows decrease of both mRNA and protein expression, but nothing about translocation.*

Thank you for your valuable comments. The research results of Darweesh et al. showed that Zc3h11a protein plays a role in regulation of NF-κB signal transduction. Depletion of Zc3h11a resulted in enhanced NF-κB mediated signaling, with upregulation of numerous innate immune related mRNAs, including IL-6 and a large group of interferon-stimulated genes. IL-6 upregulation in the absence of the Zc3h11a protein correlated with an increased NF-κB transcription factor binding to the IL-6 promoter and decreased IL-6 mRNA decay. The enhanced NF-κB signaling pathway in *Zc3h11a* deficient cells correlated with a defect in IκBα inhibitory mRNA and protein accumulation. Upon *Zc3h11a* depletion The IκBα mRNA was retained in the cell nucleus resulting in failure to maintain normal levels of the cytoplasmic IκBα mRNA and protein that is essential for its inhibitory feedback loop on NF-κB activity. These findings demonstrate that ZC3H11A can regulate the NF-κB pathway by controlling the translocation of IκBα mRNA, a mechanism that was indeed elucidated by Darweesh et al. We sincerely apologize for any lack of clarity in our original description and have now inserted the appropriate reference in the relevant section of the main text.

We deeply appreciate your valuable comments in identifying this ambiguity in our manuscript, which have significantly improved the accuracy and clarity of our work.

> *Line 283: what do you mean "may confer embryonic lethality"? Were they embryonic lethal or not?*

We sincerely appreciate your critical request for clarification. Our experimental data from 15 pregnancies of *Zc3h11a* Het-KO mice intercrosses (n = 15 litters) conclusively confirmed the absence of homozygous knockout (Homo-KO) pups at birth. These findings align with the embryonic lethality of *Zc3h11a* homozygous deletion as reported by Younis et al. We fully acknowledge the ambiguity in our original phrasing and have revised the text to:“Second, *Zc3h11a* homozygous KO (Homo-KO) mice were not obtained in our study because homozygous deletion of exons confer embryonic lethality.”Your vigilance in ensuring terminological precision has greatly strengthened the rigor of our manuscript. We hope this clarification fully resolves your concerns.

> *Line 338: What is meant that Het-KO mice were constructed at 4 weeks of age? Do these mice not have a germline mutation?*

Thank you for your valuable comments. We have revised the following content: “The germline heterozygous *Zc3h11a* knockout (Het-KO) mice were generated by CRISPR/Cas9-mediated gene editing at the embryonic stage on a C57BL/6J background, provided by GemPharmatech Co., Ltd (Nanjing, China). Phenotypic analyses were initiated when the mice reached four weeks of age.”

> *Line 346-347: how many mice were excluded due to having small eyes or ocular lesions? The methods section should state how refraction and ocular biometrics were measured.*

Thank you for your valuable comments. We have added or revised the following content: “To exclude potential confounding effects of spontaneous ocular developmental abnormalities, a total of 7 mice (4 Het-KO and 3 WT) with small eyes or ocular lesions were excluded from the observation cohort. These anomalies were consistent with the baseline incidence of spontaneous malformations observed in historical colony data of wild-type C57BL/6J mice (approximately 11%), and were not attributed to the *Zc3h11a* heterozygous knockout.

The methods for measuring refraction and ocular biometrics are as follows and have been added to the original method. Refractive measurements were performed by a researcher blinded to the genotypes. Briefly, in a darkroom, mice were gently restrained by tail-holding on a platform facing an eccentric infrared retinoscope (EIR) (Schaeffel et al., 2004; Zhou et al., 2008a). The operator swiftly aligned the mouse position to obtain crisp Purkinje images centered on the pupil using detection software (Schaeffel et al., 2004), enabling axial measurements of refractive state and pupil size. Three repeated measurements per eye were averaged for analysis. The anterior chamber (AC) depth, lens thickness, vitreous chamber (VC) depth, and axial length (AL) of the eye were measured by real-time optical coherence tomography (a custom built OCT) (Zhou et al., 2008b). In simple terms, after anesthesia, each mouse was placed in a cylindrical holder on a positioning stage in front of the optical scanning probe. A video monitoring system was used to observe the eyes during the process. Additionally, by detecting the specular reflection on the corneal apex and the posterior lens apex in the two dimensional OCT image, the optical axis of the mouse eye was aligned with the axis of the probe. Eye dimensions were determined by moving the focal plane with a stepper motor and recording the distance between the interfaces of the eyes. Then, using the designed MATLAB software and appropriate refractive indices, the recorded optical path length was converted into geometric path length. Each eye was scanned three times, and the average value was taken."

> *Line 428: what age retinas*

Thank you for your meticulous attention to the experimental design details. Regarding the age of retinal samples, we have clarified the following in the revised manuscript:" Retinas were harvested from four-week-old mice for RNA sequencing." This revision enhances the transparency and reproducibility of our methodology. We deeply appreciate your rigorous review.

> *Figure 3 D-F: these images are too small to adequately assess, please show at higher magnification. Are there fewer bipolar cells, or just decreased expression of PKC? From these images, expression of ZC3H11A does not appear decreased, but the retina appears thinner. Is that true, or are these poorly matched sections?*

Thank you for your professional insights regarding image quality and data interpretation. Your rigorous review has significantly enhanced the scientific rigor of our study. We hereby address your concerns point by point: The images in Figures 3D-F were acquired using a Zeiss LSM880 confocal microscope with a 10x eyepiece and 20x objective lens, a standard magnification for retinal section imaging that balances cellular resolution with full-thickness structural preservation. We quantified PKCα immunofluorescence intensity (a bipolar cell-specific marker) to assess changes in bipolar cell populations, rather than direct cell counting. This metric reflects PKCα expression abundance as a proxy for bipolar cell alterations (Figure 3H). To clarify terminology, we have revised the text to "bipolar cell-labelled protein PKCα immunofluorescence abundance" and detailed the methodology in the revised Methods section. Recognizing the semi-quantitative nature of fluorescence intensity analysis, we supplemented these data with Western blot results confirming reduced PKCα protein levels (Figure 3I). Zc3h11a expression was validated both by immunofluorescence intensity (Figure 3G) and Western blot (Figures 6F, H) quantification, confirming reduced expression in *Zc3h11a* Het-KO retinas. The apparent "retinal thinning" observed in histology sections stems from technical artifacts during tissue processing (fixation, dehydration, sectioning), not biological differences. HE staining, which better preserves sample morphology, showed no structural or thickness differences between Zc3h11a Het-KO mice and wild-type mice (Supplementary Figure 2).

Your expert feedback has driven us to establish a more robust validation framework. We believe the revised data now more accurately reflect the biological reality and sincerely hope these improvements meet your approval.

> *Figure 3G-J: Relative fluorescence intensity of immunohistochemistry is not a valid measure of protein expression.*

We sincerely appreciate your thorough review and valuable comments regarding the immunofluorescence quantification method in Figures 3G-J. In response to your concern that "relative fluorescence intensity is not an effective quantitative measure of protein expression," we have implemented the following improvements to our analysis and validation: To ensure result reliability, all immunofluorescence experiments followed strict protocols: experimental and control samples were fixed, stained, and imaged in the same batch to eliminate inter-batch variability. Imaging was performed using a Zeiss LSM 880 confocal microscope with identical parameters, and the relative fluorescence intensity of specific signals per unit area was measured and statistically analyzed using ZEN software. We fully acknowledge the semi-quantitative nature of relative fluorescence intensity measurements. Therefore, we validated key differentially expressed proteins using Western blot analysis: The Western blot results for Zc3h11a (Figures 6F, H) were completely consistent with the relative fluorescence intensity trends (Figure 3G). Additionally, the newly included Western blot data for PKCα (Figure 3 I) further confirmed the reliability of our relative fluorescence intensity quantification. Your expert advice has significantly enhanced the rigor of our study. Should any additional data or clarification be required, we would be pleased to provide further support.

> *Figure 4: what are the arrows pointing at? This should be in the Figure legend. What is MB? Why are there no scale bars? What is difference between E and F, not clear from legend.*

We sincerely appreciate your thorough review of Figure 4 and your valuable suggestions. In response to your concerns, we have carefully examined and improved the relevant content with the following modifications and clarifications: We sincerely apologize for not clearly indicating the arrow annotations in the original figure legend. In the revised version, we have provided detailed explanations for the arrow indicators: black arrows indicate perinuclear space dilation, blue arrows indicate cytoplasmic edema, and red arrows indicate disorganized and loosely arranged membrane discs. The updated legend has been clearly marked below Figure 4 in the main text. MB represents membrane discs, which are critical subcellular structures in the outer segments of retinal photoreceptor cells (rods and cones). They are responsible for light signal capture and transduction (containing visual pigments such as rhodopsin). The structural integrity of MB is essential for normal visual function. The scale bars in the original figures were located in the lower right corner of each subpanel, with specific parameters as follows: Figures 4A and B: magnification ×1000, scale bar 10 μm; Figures 4C and D: magnification ×700, scale bar 20 μm; Figures 4E and G: magnification ×2000, scale bar 5 μm; Figures 4F and H: magnification ×7000, scale bar 2 μm. Both Figures 4E and 4F show electron microscopy images of membrane discs (MB) in wild-type mouse photoreceptor cells. The only difference lies in the magnification: Figure 4E (×2000) demonstrates the overall arrangement pattern of membrane discs, while Figure 4F (×7000) focuses on ultrastructural details of the membrane discs (such as structural integrity). We have thoroughly checked the consistency between the figures and text, and have supplemented detailed legend descriptions in the main text. Once again, we sincerely appreciate your rigorous review, which has significantly enhanced the scientific rigor and readability of our study. Should you have any further suggestions, we would be happy to incorporate them.

> *Figure 5A: Why such a large y-axis? Figure legend does not match figure*

We sincerely appreciate your careful review of Figure 5A and your valuable suggestions regarding the figure details. In response to your concerns, we have thoroughly examined and improved the relevant content as follows: The Y-axis of the volcano plot represents $-\log_{10}$(p-value), where the magnitude of the values reflects statistical significance. Our RNA-seq data underwent rigorous multiple testing correction, and the adjusted p-values for some genes were extremely small, resulting in large values after $-\log_{10}$ transformation. We have re-examined the data distribution and confirmed that the expanded Y-axis range is solely due to a small number of highly significant genes (as shown in the figure, the majority of genes remain clustered in the lower half of the Y-axis). This result accurately reflects the true data characteristics.

We sincerely apologize for the inadvertent error in the original labeling of "Up/Down" in the figure legend. This has now been corrected, and we strictly adhere to the following threshold criteria: Significantly upregulated (Up): adjusted p-value < 0.05 and $\log_2$(FC) ≥ 1. Significantly downregulated (Down): adjusted p-value < 0.05 and $\log_2$(FC) ≤ -1. To ensure the reliability of our conclusions, we have rechecked the raw data, statistical analysis, and visualization process. We confirmed that all significant genes strictly meet the above threshold criteria and that the visualization accurately reflects the true results. The revised figure has been updated in the manuscript as Figure 5A. We deeply appreciate your valuable feedback, which has helped us correct the errors in the figure and improve its accuracy and readability.

> *Figure 6F: Based on the western blot, only Zc3h11a appears different.*

Thank you for your careful evaluation of the Western blot data in Figure 6F. We fully understand your concerns regarding the visual differences in PI3K and p-AKT/AKT bands and appreciate the opportunity to clarify the quantitative methodology and biological significance of these findings. Below we provide a detailed explanation of the experimental design and data analysis.

First, the data for each group were derived from retinal samples of three independent mice, with all experiments performed in parallel to control for technical variability. Image analysis was conducted using ImageJ software with standardized settings for grayscale quantification. Zc3h11a and PI3K levels were normalized to GAPDH as an internal reference, while p-AKT levels were calculated as a ratio to total AKT. The results showed that Zc3h11a protein levels were significantly reduced ($p < 0.01$, Figures 6F and H), consistent with the expected effects of heterozygous knockout, with good agreement between visual and statistical results. For PI3K and p-AKT/AKT, the bands appeared visually similar due to: The nonlinear nature of Western blot chemiluminescence signals in the saturation range, which compresses subtle quantitative differences in the images; the fact that p-AKT represents only 5-15% of the total AKT pool, making small proportional changes difficult to discern visually. However, it is important to note that both PI3K and p-AKT/AKT showed statistically significant differences between groups ($p < 0.001$ and $p < 0.01$, respectively; Figures 6G and I). Furthermore, signal transduction pathways exhibit cascade amplification effects - in the PI3K-AKT pathway, even small changes in upstream proteins can produce significant downstream effects (e.g., NF-κB activation) through kinase cascades (Figure 6J). Additionally, our RNA-Seq results revealed activation of the PI3K-AKT signaling pathway in *Zc3h11a* Het-KO mice (Figure 5D), and the qRT-PCR results were consistent with the western blot results (Figure 6A-C). Your expert comments have prompted us to present these data differences with greater biological rigor. Although the visual differences are subtle, based on statistical significance, pathway characteristics, and RNA sequencing, and qRT-PCR data, we believe these changes have biological relevance. We sincerely appreciate your commitment to data rigor and respectfully request your recognition of both the experimental results and the scientific logic of this study.

> *Figure 8: What is the role of ZC3H11A in this figure? Are the authors proposing that ZC3H11A regulates the translation of IκBα? They have not shown any evidence of that.*

Thank you for your insightful exploration of the role of ZC3H11A in Figure 8. We appreciate your critical review and hope to elucidate the mechanistic framework behind our findings. In Figure 8, *Zc3h11a* is depicted as a regulator of IκBα mRNA nucleocytoplasmic transport, a mechanism originally elucidated by Darweesh et al. Their studies demonstrated that *Zc3h11a* binds to IκBα mRNA and promotes its nuclear export. Loss of *Zc3h11a* results in nuclear retention of IκBα mRNA, leading to reduced cytoplasmic IκBα protein levels and subsequent hyperactivation of the NF-κB pathway. While the specific nuclear export mechanism has been elucidated by Darweesh et al., our study demonstrates that *Zc3h11a* haploinsufficiency results in decreased IκBα mRNA and protein levels in the retina (Figure 7), linking *Zc3h11a* haploinsufficiency to NF-κB pathway dysregulation in myopia and highlighting that these molecular insights can be extended to a new pathological context (myopia). Your critical comments have enhanced the clarity of our mechanistic concepts and we hope that these descriptions will demonstrate the importance of *ZC3H11A* as a new candidate gene for myopia.

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